

SERBP1 interacts with PARP1 and is present in PARylation-dependent protein complexes regulating splicing, cell division, and ribosome biogenesis

Reviewed Preprint

v2 • January 6, 2025

Revised by authors

Reviewed Preprint

v1 • July 2, 2024

Kira Breunig, Xiufen Lei, Mauro Montalbano, Gabriela DA Guardia, Shiva Ostadrahimi, Victoria Alers, Adam Kosti, Jennifer Chiou, Nicole Klein, Corina Vinarov, Lily Wang, Mujia Li, Weidan Song, W Lee Kraus, David S Libich, Stefano Tiziani, Susan T Weintraub, Pedro AF Galante, Luiz OF Penalva 

Children's Cancer Research Institute, UT Health San Antonio, San Antonio, USA • Mitchell Center for Neurodegenerative Diseases, University of Texas Medical Branch, Galveston, USA • Department of Neurology, University of Texas Medical Branch, Galveston, USA • Centro de Oncologia Molecular, Hospital Sírio-Libanês, São Paulo, Brazil • Department of Cell Systems and Anatomy, UT Health San Antonio, San Antonio, USA • Department of Biochemistry and Structural Biology, UT Health San Antonio, San Antonio, USA • Department of Nutritional Sciences, College of Natural Sciences, University of Texas at Austin, Austin, USA • Laboratory of Signaling and Gene Regulation, Cecil H. and Ida Green Center for Reproductive Biology Sciences, The University of Texas Southwestern Medical Center, Dallas, USA • Department of Pediatrics, Dell Medical School, University of Texas at Austin, Austin, USA • Department of Oncology, Dell Medical School, University of Texas at Austin, Austin, USA

 https://en.wikipedia.org/wiki/Open_access

 Copyright information

eLife Assessment

This study reports **valuable** insights into the interactome of the RNA-binding protein SERBP1 and possible links through PARylation to diverse processes, including splicing, cell division, and ribosome biogenesis. The diversity of processes SERBP1 may regulate means this work would be of very broad interest to the cell biology community. The proteomics data are **solid**, but the functional connection to downstream processes and the link to Alzheimer's disease, while **compelling**, still require further examination. These latter data currently rely on a very limited set of experiments and patient samples with questionable quality of preservation and methodology.

<https://doi.org/10.7554/eLife.98152.2.sa3>

Abstract

RNA binding proteins (RBPs) containing intrinsically disordered regions (IDRs) are present in diverse molecular complexes where they function as dynamic regulators. Their characteristics promote liquid-liquid phase separation (LLPS) and the formation of membraneless organelles such as stress granules and nucleoli. IDR-RBPs are particularly relevant in the nervous system and their dysfunction is associated with neurodegenerative diseases and brain tumor development. Serpine1 mRNA-binding protein 1 (SERBP1) is a unique member of this group, being mostly disordered and lacking canonical RNA-binding

domains. We defined SERBP1's interactome, uncovered novel roles in splicing, cell division and ribosomal biogenesis, and showed its participation in pathological stress granules and Tau aggregates in Alzheimer's brains. SERBP1 preferentially interacts with other G-quadruplex (G4) binders, implicated in different stages of gene expression, suggesting that G4 binding is a critical component of SERBP1 function in different settings. Similarly, we identified important associations between SERBP1 and PARP1/polyADP-ribosylation (PARylation). SERBP1 interacts with PARP1 and its associated factors and influences PARylation. Moreover, protein complexes in which SERBP1 participates contain mostly PARylated proteins and PAR binders. Based on these results, we propose a feedback regulatory model in which SERBP1 influences PARP1 function and PARylation, while PARylation modulates SERBP1 functions and participation in regulatory complexes.

Introduction

RNA binding proteins (RBPs) are a diverse group of regulators with 1,500 members cataloged in the human genome (1, 2). They display a variety of RNA binding domains (RRMs, zinc-fingers, KH domains, dsRBD, etc.) and control multiple stages of gene expression, including RNA processing (splicing, capping, 3' end processing, and polyadenylation), modification, transport, decay, localization, and translation (2, 3). RBPs are particularly relevant in the nervous system, where they regulate critical aspects of neurogenesis, neuronal function, and nervous system development (4, 5). Alterations in RBP expression and function are linked to neurological disorders, neurodegenerative diseases, and brain tumor development (6–12).

RBPs assemble in complexes to regulate gene expression. RBP interactions can take place in the context of macromolecular complexes associated with liquid-liquid phase separation (LLPS). This process supports important regulatory environments and is the foundation for membraneless organelles such as stress granules, Cajal bodies, nucleoli, and P-bodies (13–15). RBPs in LLPS tend to share characteristics that include intrinsically disordered regions (IDRs) and RGG boxes. IDR-RBPs can interact with RNA without the presence of specific folded RNA-binding domains (16). These features provide unique structural and biophysical characteristics to dynamically regulate translation, stress response, RNA processing, and neuronal function (17).

Many RBPs in LLPS share a binding preference for G-quadruplexes (G4s). G4s are complex DNA or RNA structures that display stacked tetrads of guanines stabilized by Hoogsteen base pairing. They function as regulatory elements in different stages of gene expression, promote LLPS, and are implicated in cancer, viral replication, neurodegenerative diseases, neurological disorders, and prion diseases (18–21).

RBPs containing IDRs and RGG motifs are particularly relevant in the nervous system. Their misfolding contributes to the formation of pathological protein aggregates in Alzheimer's disease (AD), Frontotemporal Lobar Degeneration (FTLD), Amyotrophic Lateral Sclerosis (ALS), and Parkinson's disease (PD) while their aberrant expression has been linked to glioblastoma development (22–24). SERBP1 (Serpine1 mRNA-binding protein 1) is a unique member of this group of RBPs for being mostly disordered and lacking typical RNA binding domains other than two short RG/RGG motifs (25). SERBP1 is a highly conserved protein with homologs identified in yeast, plants, invertebrates, and vertebrates. SERBP1 is highly expressed in neuronal and glioma stem cells and a decrease in its levels is required for neuronal differentiation (26). In glioblastoma, SERBP1 functions as a central regulator of metabolic pathways, coordinating the expression of related enzymes and associated factors implicated in serine biosynthesis, one-carbon, and 5'-methylthioadenosine (MTA) cycles. Ultimately, SERBP1 impacts methionine production and, consequently histone methylation and the expression of genes implicated in neuronal differentiation (26).

In the present study, we investigated the SERBP1 interactome. Overall, our results established SERBP1 as a multi-regulatory protein, corroborating that RBPs containing IDRs are dynamic and can assemble different complexes. Despite its predominant cytoplasmic localization, SERBP1 is associated with proteins in specific nuclear complexes implicated in splicing regulation, cell division, and ribosome biogenesis. Approximately a third of SERBP1-associated proteins bind to G4s. These proteins are implicated in various stages of gene expression, indicating that G4s function as SERBP1 regulatory motifs in different scenarios. Finally, our study has identified important associations between SERBP1 and PARP1 that suggest a regulatory loop, in which SERBP1 influences PARylation, and PARP activity. At the same time, PARylation of SERBP1 and its partner proteins modulate their association and function.

Results

SERBP1 interactome reveals its associations with diverse regulatory complexes

To identify SERBP1-associated proteins and molecular complexes in which it is involved, we conducted pull down experiments in 293T cells followed by proteomics analysis. Cells were transfected with pSBP-SERBP1 or pUltra-SERBP1(control). The SBP tag is a short peptide with high affinity for streptavidin. Protein complexes are then isolated via pulldown using streptavidin beads. Analysis by mass spectrometry yielded a high-confidence list of 570 SERBP1-associated proteins. Peptides for these proteins were only present in the pSBP-SERBP1 pull-down samples or displayed a minimum of 3-fold enrichment in comparison to control – **Table S1**, **Figure S1**. Gene Ontology (GO) analyses ([27](#)–[29](#)) of identified partners indicated that SERBP1 participates in diverse regulatory complexes in the cytoplasm and nucleus – **Figure 1A-B** and **Table S1**. We compared our data to the results of two large-scale studies that used proximity-dependent biotinylation (Bio ID) ([30](#), [31](#)) to identify protein-protein interactions and checked the datasets in BioGRID ([32](#)). 146 SERBP1 interactors identified in our analysis were also identified in other high-throughput studies – **Table S1**.

Although SERBP1's partners indicate its participation in different stages of translation as well as in its regulation, the many associations with translation initiation factors suggest that SERBP1's primary function is in initiation – **Table S1**. As expected, based on previous studies ([33](#)–[36](#)), we identified several ribosomal proteins associated with SERBP1 – **Table S1**. SERBP1 knockdown in U251 and U343 cells resulted in a decrease of overall translation as shown by puromycin incorporation assays – **Figure S2A**.

SERBP1 is mostly localized in the cytoplasm. SERBP1 nuclear localization increases in response to stress and is regulated by arginine methyltransferases ([37](#)–[39](#)). SERBP1-associated proteins suggest it participates in nuclear functions; out of 570 associated proteins, 344 display cytoplasmic/nuclear localization, while 100 are exclusively found in the nucleus – **Table S1**. GO analyses of this group identified specific nuclear complexes. The functions of SERBP1-associated proteins suggest that SERBP1 is involved in different stages of ribosome biogenesis – **Table S1**. SERBP1 participation in ribosome biogenesis is corroborated by its presence in nucleoli – **Figure S2B-C**. Results of our previous RIP-Seq analysis ([26](#)) and a CLIP-seq study ([36](#)) showed that SERBP1 binds preferentially to multiple snoRNAs (**Table S2**) and at specific sites in 18S and 28S rRNA. These results, aligned with the fact that SERBP1 associates with FBL and DKC1, two critical snoRNP components, strongly indicate SERBP1 involvement in rRNA modification. Besides ribosome biogenesis, SERBP1 is also likely involved in different stages of RNA processing, in particular splicing, and telomere maintenance – **Figure 1A-B**.

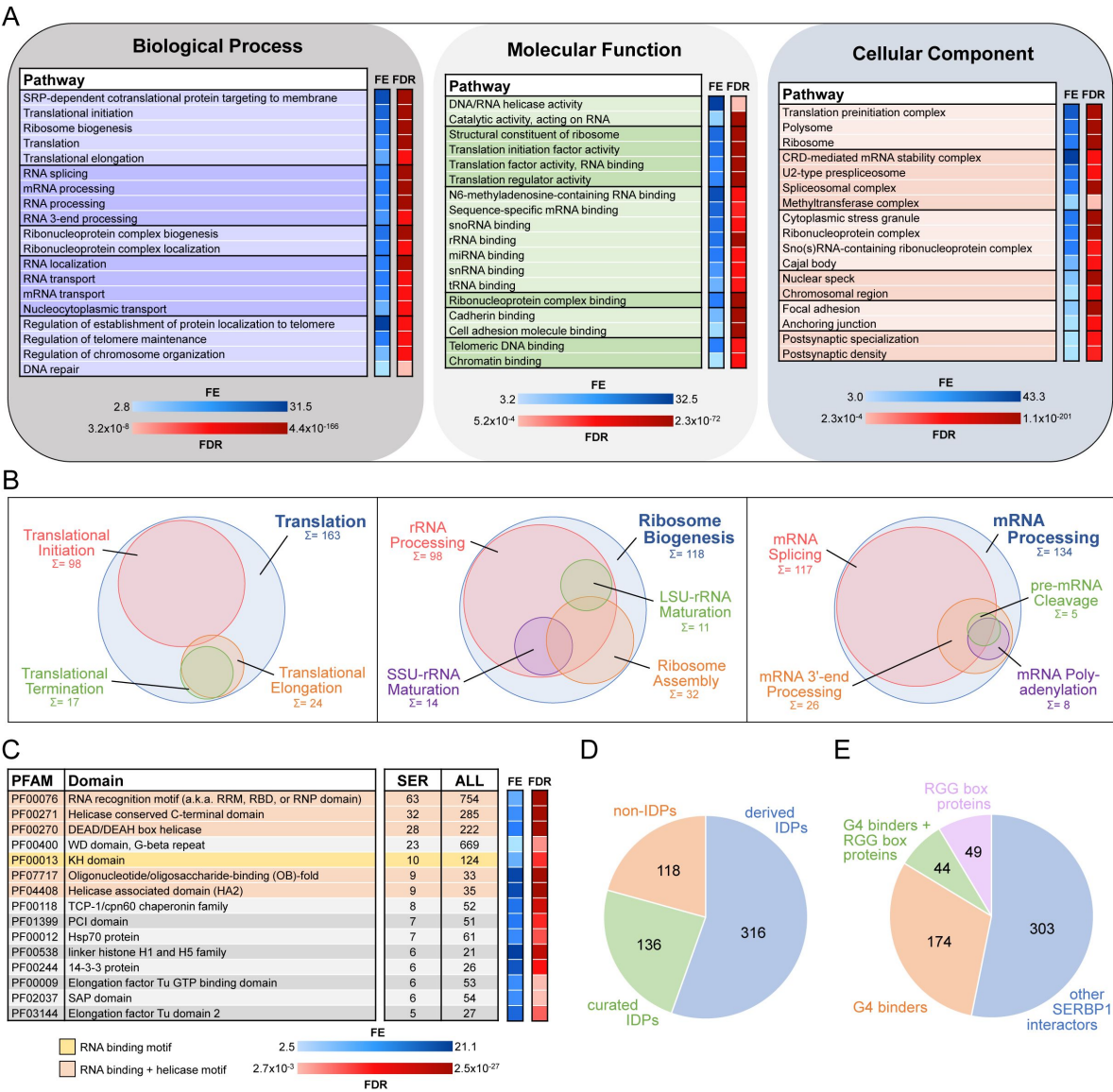


Figure 1

Gene ontology (GO) analysis and characterization of SERBP1-interacting proteins.

A) SERBP1-interacting factors were identified via pull-down analysis in 293T cells. GO enrichment analysis (Biological Processes, Molecular Function, and Cellular Component) was performed with ShinyGO (27). A selection of the most relevant GO terms is represented. FE = fold enrichment; FDR = false discovery rate. **B)** Visualization of associated GO-enriched terms in three main regulatory branches: translation, ribosome biogenesis, and RNA processing. Sum and circle size represent the count of SERBP1 interactors assigned to each GO term. **C)** Representation of top enriched protein domains according to PFAM (42) among SERBP1 interactors. Only PFAM domains with counts of 5 or more and significant fold enrichment (FE) and false discovery rate (FDR) were included. The occurrence of each domain within the total human proteome (ALL) is listed as a reference. **D)** Occurrence of intrinsically disordered proteins (IDPs) (45) among SERBP1 interactors according to PFAM ("derived" = automatic entries from databases; "curated" = manually checked). **E)** Distribution of identified SERBP1-interacting proteins in reference to the presence of RGG boxes (50) and binding to G-quadruplexes (G4s) (46, 47). Data used to generate the figures are in Tables S1, S4 and S6.

SERBP1 is a highly conserved protein. In *S. cerevisiae*, SERBP1 homolog is STM1 (40). According to BioGRID (32), 183 proteins were identified as STM1 interactors. Of the 144 identified human homologs, we found that 53 were SERBP1 interactors – **Table S3**. GO analyses of STM1 interactors indicated that STM1 is also a multi-functional protein. Comparisons of enriched GO terms from the SERBP1 and STM1 analyses suggest that several regulatory functions assigned to SERBP1 based on its associated proteins are evolutionarily conserved, including translation, ribosome biogenesis, RNA processing and telomere maintenance – **Table S3**. SERBP1 also has homologs in plants. A recent study in *A. thaliana* established that AtRGG proteins (SERBP1 homologs) interact with ribosomal proteins and proteins involved in RNA processing and transport (41).

SERBP1 preferentially associates with helicases and G-quadruplex binders

In order to define the characteristics of SERBP1-associated proteins, we cataloged them according to their protein domains using the Pfam database (42) and performed an enrichment analysis. RNA binding and helicase are the two domain families that appeared most often – **Figure 1C**, **Figure S3**, and **Table S4**. Based on the results of the proteomics analyses and gene expression correlation studies, DDX21, DHX9, and DHX15 are SERBP1's top-associated helicases – **Table S5**.

SERBP1 displays large intrinsically disordered regions (IDRs) (25). IDRs are ubiquitous within biomolecular condensates in the cytoplasm and nucleus created via liquid-liquid phase separation (LLPS). They lead to the formation of membraneless organelles such as stress granules, P-bodies, and nucleoli (43, 44). Roughly 80% of SERBP1-associated proteins contain IDRs (45) and are present in different membraneless organelles, as discussed below – **Figure 1D** and **Table S4**.

SERBP1 is a G-quadruplex (G4) binder (26, 46), as are 38% of its associated proteins (46, 47) – **Table S6**. RBPs, including SERBP1, often use RGG boxes to bind G4 motifs (48, 49). We found that among 93 SERBP1-associated proteins containing RGG boxes (50), 44 are G4 binders – **Figure 1E** and **Table S6**. G4 binders associated with SERBP1 are mostly proteins implicated in translation, mRNA stability, and splicing, indicating that SERBP1 regulation via G4 binding takes place in different scenarios. Among G4 binders, we identified several RNA helicases (e. g. DDX5, DDX21, DDX3X, and DHX9) that enhance translation by unfolding G4s in 5' UTRs (51–54) – **Figure S3**, **Tables S1** and **S6**. Since SERBP1 lacks helicase domains, interaction with these proteins could be critical for SERBP1 function in translation.

G4s regulate different aspects of the life cycles of RNA viruses (55, 56). GO terms related to viral translation are among those with the highest fold enrichment in our analyses – **Table S1**. In a study to define the SARS-CoV-2 RNA interactome, 168 human proteins were identified (after exclusion of ribosomal proteins and EI3F factors) (57), including SERBP1 and 92 of its associated proteins – **Table S6**. Of those 92 proteins, 52 are G4 binders – **Table S6**.

SERBP1 affects cell division

SERBP1 has been previously implicated in chromosome segregation and shown to localize in M bodies during mitosis (36). In agreement, GO analyses of SERBP1-associated proteins revealed chromosome segregation as well as several related functions as enriched terms. Examination of proteins linked to these terms indicated that SERBP1 interacts with proteins in chromosomes as well as centrosomes and mitotic spindles – **Figure 2A** and **Table S1**. To evaluate SERBP1's impact on cell division, U251 and U343 SERBP1 knockdown and control cells were exposed to 20 nM of paclitaxel for 24 hours and later stained with α -tubulin and DAPI. In both cases, SERBP1 knockdown cells had markedly increased numbers of polynucleated cells, reflecting mitotic catastrophe – **Figure 2B-D**. SERBP1 involvement in cell division appears to be conserved. STM1, the SERBP1 yeast homolog, is a suppressor of mutations in genes regulating mitosis (58). Confirming that report, our GO analyses of STM1 genetic interactors showed enrichment for cell division – **Table S3**.

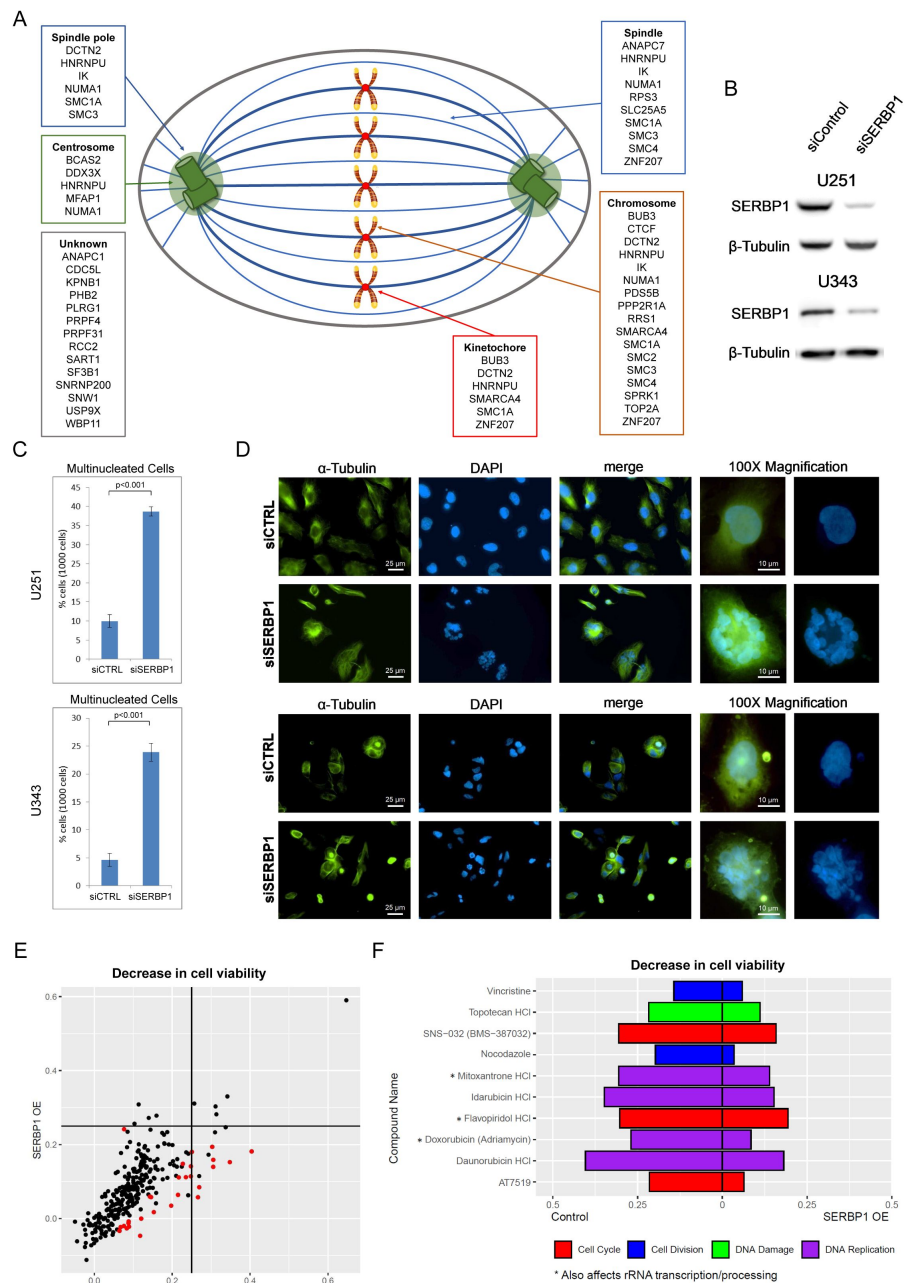


Figure 2

SERBP1 is implicated in cell division.

A) SERBP1-associated proteins and their presence in specific cellular structures relevant to mitosis. **B)** Western blot showing SERBP1 knockdown in U251 and U343 cells. **C)** Multinucleated cells in U251 and U343 control and SERBP1 knockdown (KD) cells after treatment with paclitaxel, an agent that causes cell cycle arrest. SERBP1 KD increased the number of multinucleated nuclei. Data are shown as means of counts of 1000 cells $\pm$ standard deviation (in triplicate) and statistical significance was determined by Student's t-test. $P < 0.001$. **D)** Aspect of cells exposed to paclitaxel. On the left staining with anti- α -Tubulin; in the middle staining with DAPI showing an increased number of multinucleated cells after siSERBP1 KD. 100X magnification for detailed visualization of a single multinucleated cell. **E)** Plot shows the results of a cell viability screening (291 drugs) performed in U343 control vs. U343 SERBP1 over-expressing (OE) cells. Red dots correspond to drugs whose impact on cell viability was significantly different between U343 control and U343 SERBP1 OE. **F)** Highlights of the screening showing cell cycle/division and DNA damage/replication inhibitors whose impact on cell viability was smaller in U343 SERBP1 OE in comparison to control. Datasets used to prepare the figure and detailed analysis are in Table S1 and S7.

We conducted a viability screening with 269 compounds in GBM U343 and U343 over-expressing (OE) SERBP1 (26). Corroborating SERBP1 involvement in cell division, we determined that SERBP1 OE cells are less sensitive to drugs affecting cell division, cell cycle, and DNA replication. Interestingly, three of these drugs (Mitoxantrone, Flavopiridol, and Doxorubicin) are also known to affect rRNA transcription and processing – **Figure 2E-F** and **Table S7**.

SERBP1 interacts with splicing factors and influences splice site selection

Among identified SERBP1 partners, 134 proteins (23%) are implicated in mRNA processing, and from this group, 117 participate in mRNA splicing. They include many well-known alternative splicing regulators such as hnRNPU, KHDRBS1, PTBP1, SRSF6, TRA2B, SRSF2, and hnRNPA1 – **Figure 3A**. In particular, hnRNPU, TRA2B, and KHDRBS1 display high expression correlation with SERBP1 in multiple scenarios – **Table S5**. To determine if SERBP1 expression levels influence splicing, we conducted an RNAseq study in U251 cells. SERBP1 knockdown altered the splicing of 1,625 events ($PSI > 0.1$, $FRD < 0.05$) with 70% of them being exon skipping events – **Figure 3B** and **Table S8**. Next, we examined a SERBP1 CLIPseq dataset (36) to search for evidence of SERBP1 direct participation in regulated splicing. From the 1,625 splicing events altered in SERBP1 knockdown cells, CLIP sites were observed close to pertinent splice sites (< 100 nucleotides) of 443 (27%) of them – **Figure 3C** and **Table S8**. SERBP1's association with many known alternative splicing regulators suggests that its impact on regulated splicing requires interaction with other factors. Since hnRNPU was identified as one of SERBP1's main partners and they display strong expression correlation in multiple instances (**Figure 3D**), we decided to conduct additional RNAseq analysis in U251 cells to determine if the two proteins co-regulate splicing events. Circa 30% of the events identified in SERBP1 knockdown cells were also observed upon hnRNPU knockdown, with changes in the same direction – **Figure 3E-F** and **Table S8**.

The results of the splicing analysis also revealed another potential route by which SERBP1 could affect ribosome biogenesis. Most expressed snoRNAs are embedded in introns of protein-coding genes and lncRNAs. Our analysis indicates that SERBP1 knockdown affects splicing events that include introns containing snoRNAs in 13 host genes – **Table S8**.

SERBP1 has a two-way association with PARP1 and PARylation

We identified important associations between SERBP1 and PARP1, an enzyme that catalyzes the addition of poly(ADP-ribose) (PAR) on target proteins – **Figure 4A**. PARP1 is implicated in DNA repair and regulates multiple stages of gene expression, including transcription, splicing, ribosome biogenesis, and translation (59–61). SERBP1 and PARP1 showed strong expression correlated in several scenarios and similar expression profiles during cortex development – **Figure 4B-C**, **Table S5**.

Results of high throughput studies established that SERBP1 gets PARylated and binds to PAR (62–64) – **Figure 4D**, **Table S6**. SERBP1 PARylation by PARP1 was confirmed in an *in vitro* assay – **Figure 4E**. Moreover, we determined that circa 56% of all identified SERBP1 interactors get PARylated and/or bind PAR (**Figure 4F**, **Table S6**), suggesting that PARylation is an important component in the assembly of protein complexes in which SERBP1 participates. We further investigated SERBP1-PARP1 interaction in 293T and U251 cells in the context of PARP activation. Cell exposure to H_2O_2 enhances PARP1 activity and protein PARylation as shown by Western blots probed with an anti-poly-ADP-ribose binding reagent – **Figure 4G**. In U251 cells, H_2O_2 treatment caused SERBP1 to shift to the nucleus, where it co-localized with PARP1 – **Figure S4A**. Next, we co-transfected SBP-SERBP1-His and Flag-PARP1 in 293T cells and conducted pull-down experiments with streptavidin beads using extracts from cells treated with H_2O_2 and control. An increase in SERBP1-PARP1 interaction was observed in cells treated with H_2O_2 – **Figure 4H**. On the other hand, treatment with the PARP inhibitor PJ34 decreased SERBP1 interaction with PARP1 and G3BP1 as indicated by the results of the pulldown analysis with streptavidin beads – **Figure 4I**.

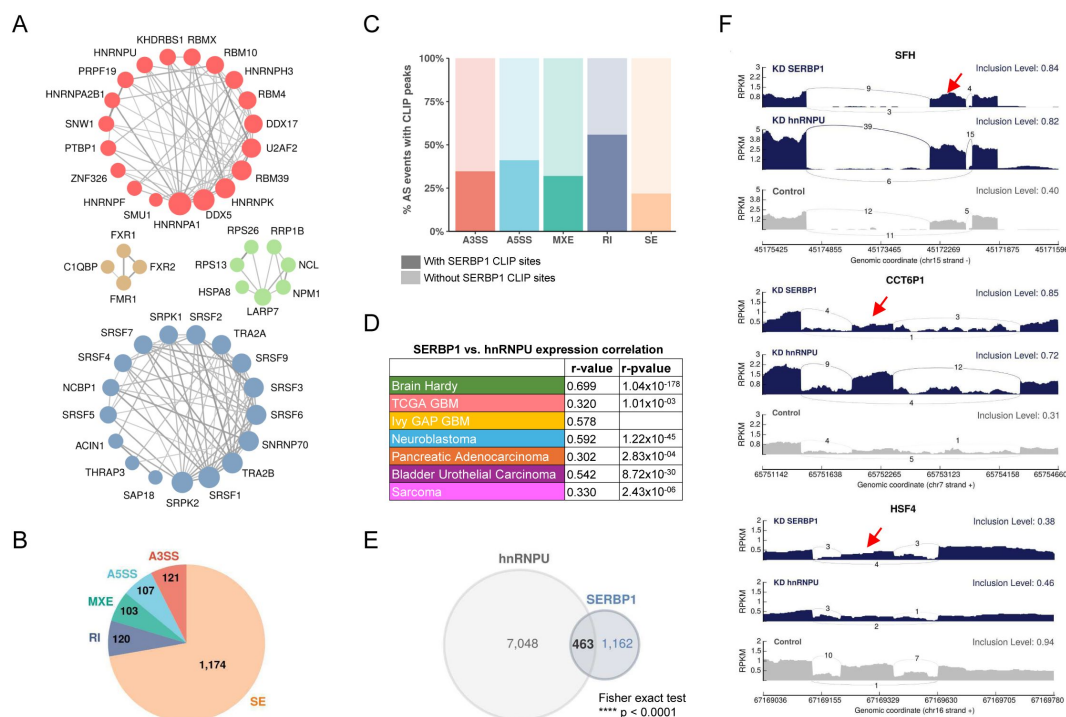


Figure 3

SERBP1 influences splicing.

A) SERBP1-interacting factors are preferentially associated with splicing. Networks show connections between splicing factors identified as SERBP1 interactors. **B)** SERBP1 knockdown in U251 cells produced major changes in splicing. Pie chart displays the distribution of splicing events affected by SERBP1 knockdown according to their type: SE, exon skipping; RI, intron retention; MXE, multiple exclusive exons; A5SS, alternative 5' splice sites; A3SS, alternative 3' splice site. **C)** Bar graphs showing the percentage of splicing events affected by SERBP1 knockdown with evidence of SERBP1 binding sites close (<100nt) to regulated splice sites. **D)** hnRNP was identified as a potential partner of SERBP1 in splicing regulation. SERBP1 and hnRNP display high expression correlation in normal and tumor tissues. **E)** Venn diagram shows a strong overlap between the splicing events affected by SERBP1 and hnRNP knockdowns in U251 cells. **F)** Sashimi plots showing examples of splicing events affected by both SERBP1 and hnRNP knockdowns in U251 cells. The red arrows indicate affected exons. Datasets used to prepare the figure and detailed analyses are in Table S8.

Transgenic SERBP1 expression in 293T cells increased PARylation, as revealed by western blot probed with PAR detecting reagent – **Figure 4J** [64](#). To determine if PARylation/PAR binding affects SERBP1 interactions, we transfected 293T cells with an SBP-SERBP1 expressing vector. The experimental group was treated later with PJ34. SERBP1-associated proteins were isolated via pulldown with streptavidin beads and the presence of PARylated proteins was evaluated by Western blot. PARP inhibition decreased the amount of PARylated proteins associated with SERBP1 as shown in the pulldown lanes – **Figure S4B**.

A recently published screening identified genes conferring sensitivity to PARP inhibitors, Olaparib, Rucaparib, and Talazoparib ([65](#) [66](#)). From this study, we generated a list with top hitters for each drug and compiled the results to identify genes appearing in two or more analyses. The 424 identified genes include SERBP1 and 82 of its interactors – **Table S7**. In agreement, using a proliferation assay, we determined that SERBP1 partial knockdown makes U251 and U343 GBM cells more sensitive to the PARP inhibitor PJ34 – **Figure S4C**.

PARP1 and several of its characterized interactors were determined to be associated with SERBP1 – **Table S1**. Our investigation has determined that the list of PARP1-SERBP1 shared factors is in fact more extensive. A comparison between SERBP1-associated factors and PARP1 interactors detected via proximity labeling ([66](#) [67](#)) identified 199 common factors (including SERBP1 and PARP1), of which about 80% are also PARylated and/or bind PAR ([62](#) [63](#)–[64](#) [68](#)) – **Figure 5A** [69](#) and **Table S6**. These proteins are implicated in splicing regulation, ribosome biogenesis, and DNA repair – **Figure 5B** [70](#) and **Table S6**. Among shared SERBP1-PARP1 interactors that get PARylated and/or bind to PAR are multiple key factors implicated in rRNA transcription, processing, and modification, including NOLC1, NOP16, NAT10, DKC1, DDX21, NCL, SMARCA4 and TCOF1 – **Figure 5C** [71](#) and **Table S6**. The binding of snoRNAs to PARP1 stimulates its catalytic activity in the nucleolus independent of DNA damage. Activated PARP1 PARylates the RNA helicase DDX21 to promote rDNA transcription ([67](#) [72](#)). DDX21 was identified as a top SERBP1-associated factor, and expression of the two proteins was highly correlated in several scenarios – **Tables S1** and **S5**. Moreover, we and others have determined that SERBP1 preferentially binds snoRNAs, including the ones implicated in PARP1 activation: snoRA37, snoRA74A, and snoRA18 ([67](#) [73](#)). Out of 58 snoRNAs bound by PARP1 ([68](#) [74](#)), 32 of them are also targeted by SERBP1 – **Table S2**.

Adding up to SERBP1-PARP1 strong functional association, we determined that about 50% of SERBP1 interactors known to bind PAR, are also G4 binders – **Figure S5A**, **Table S6**. Different types of protein domains bind to PAR, including RGG boxes and RRM (s) ([69](#) [75](#), [76](#)). These two types of domains are also among the ones that bind G4s. About half of SERBP1-associated proteins that were identified as both PAR and G4 binders contain RGG boxes and/or RRM (s) ([50](#) [77](#), [78](#)) – **Figure S5B** and **Table S6**. Based on our findings, we propose the existence of a feedback regulatory model in which SERBP1 influences PARP1 function and PARylation, while PARylation modulates SERBP1 functions and association with its protein partners – **Figure 5D** [79](#).

SERBP1 is in association with proteins present in membraneless organelles and Tau aggregates

GO analysis of SERBP1-associated proteins based on cellular compartment ([72](#) [80](#)) determined that SERBP1 interacts with factors often present in membraneless organelles, including stress granules, P bodies, Cajal bodies, nuclear speckles, paraspeckles, and nucleoli, which suggests SERBP1's likely participation in protein complexes present in these organelles – **Table S9** and **Figure 6** [81](#). SERBP1 binding to snoRNAs and scaRNAs further supports its presence in Cajal bodies and nucleolus – **Table S2**. Liquid-liquid phase separation (LLPS) is a main contributor to the formation of membraneless organelles or granules ([73](#) [82](#)). We have previously established that SERBP1 forms biomolecular condensates *in vitro* that are modulated by RNA ([25](#) [83](#)). A characteristic of proteins in LLPS is the presence of IDRs ([74](#) [84](#), [85](#)). Of all SERBP1-associated proteins present in membraneless organelles, 77% of them contain IDRs – **Table S9**.

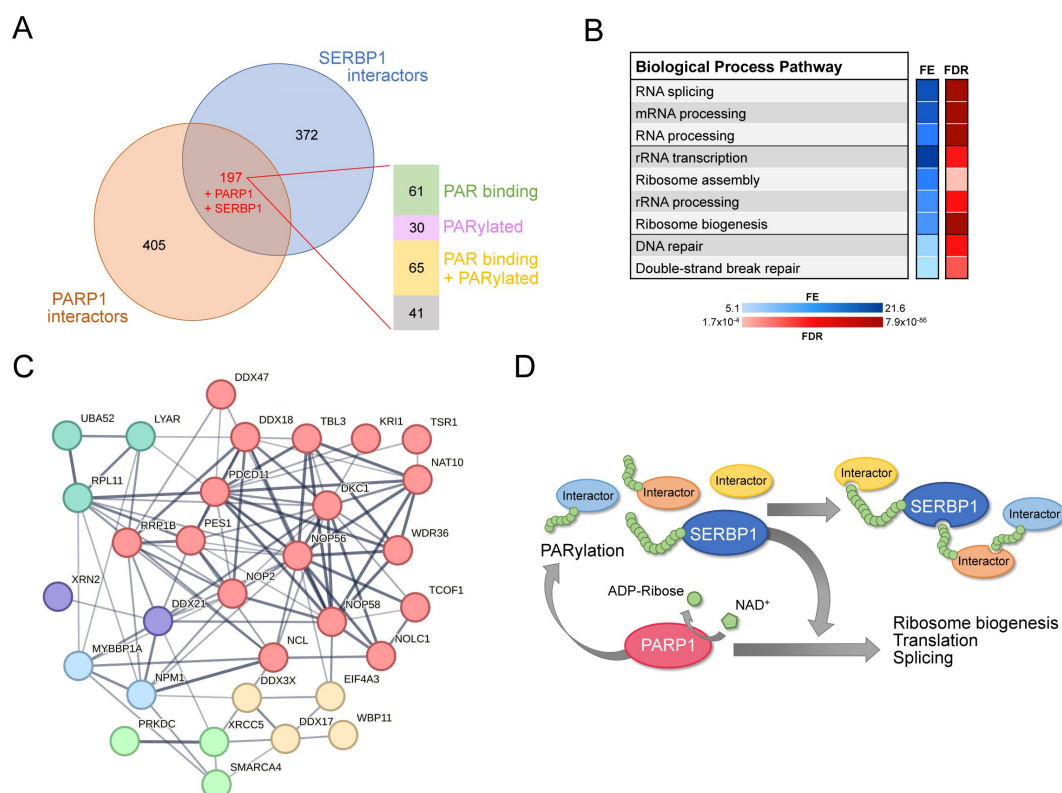


Figure 5

Shared SERBP1 and PARP1 interactors.

A) Venn diagram shows the overlap between PARP1 (66) and SERBP1 interactomes. Analysis of shared interactors indicated that the majority of them are PAR binding (64) and/or get PARylated (62, 63). **B)** Selection of enriched GO terms (biological processes) related to SERBP1-PARP1 shared interactors according to ShinyGO (27). FE = fold enrichment; FDR = false discovery rate. **C)** Network showing shared SERBP1 and PARP1 interactors implicated in ribosome biogenesis. **D)** Proposed SERBP1-PARP1 feedback model; SERBP1 function and association with partner proteins is modulated by PARylation while SERBP1 influences PARP activity. Datasets used to prepare the figure and detailed analyses are in Tables S1 and S6.

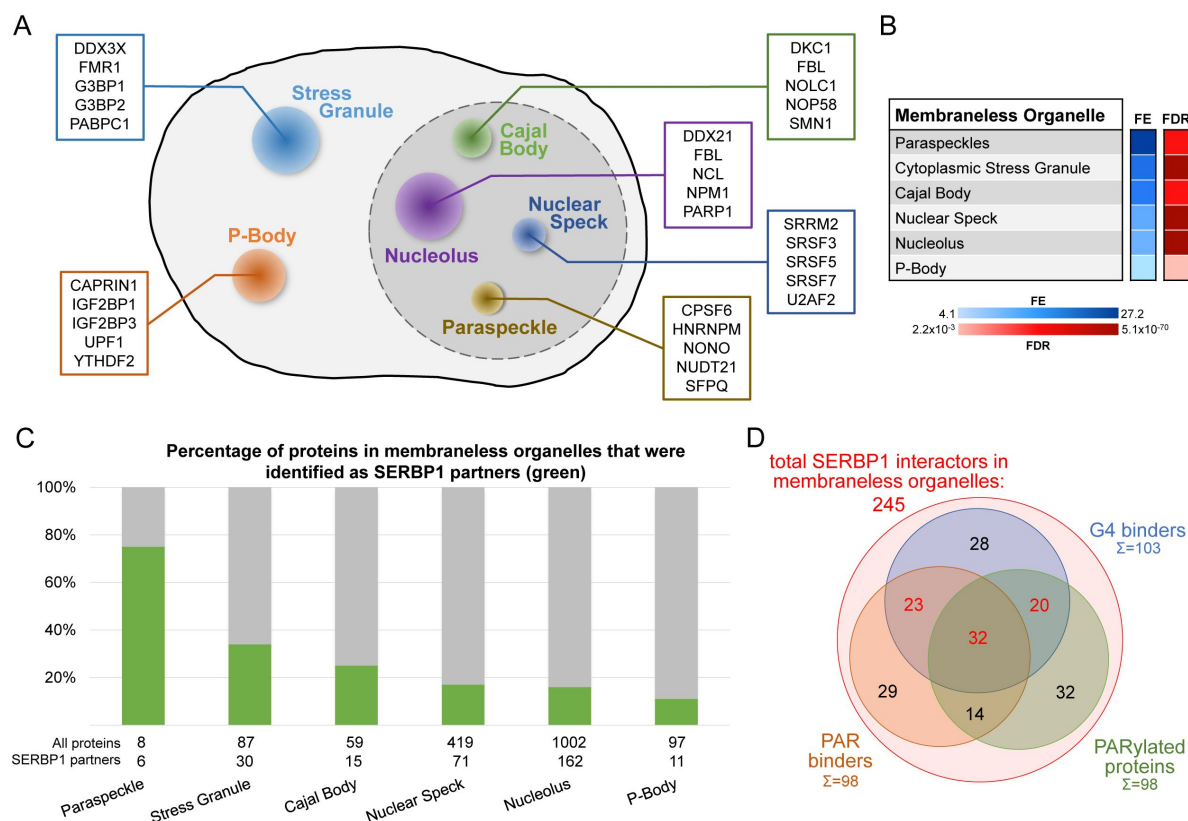


Figure 6

SERBP1-associated proteins are present in membraneless organelles.

A) Different types of membraneless organelles in a cell and examples of SERBP1-associated proteins present in each structure. **B)** Membraneless organelles identified in the GO enrichment analysis (cellular component) of SERBP1 interactors (72 [\[46\]](#)). FE = fold enrichment; FDR = false discovery rate. **C)** Bar graph showing the distribution of SERBP1-associated proteins in different membraneless organelles. **D)** Venn diagram representing the distribution of SERBP1-associated proteins present in membraneless organelles in respect to G4 binding (46 [\[46\]](#), 47 [\[47\]](#)), PAR binding (64 [\[46\]](#)), and PARylation (62 [\[46\]](#), 63 [\[47\]](#)). Datasets used to prepare the figure and detailed analyses are in Tables S1, S6 and S9.

SERBP1 has been observed in Tau aggregates in Alzheimer's disease (AD) brains and is part of a group of 261 proteins detected in several studies that examined the composition of Tau aggregates (9). We further investigated SERBP1's association with pathological Tau by co-staining SERBP1 in AD tissues with phospho-tau (Thr231), which is associated with large inclusion and paired helical filaments (PHF) (76). We found a signal overlap between large hyperphosphorylated Tau inclusions and SERBP1 puncta (**Figure 7A**), confirming their close interaction in AD tissue. The interaction between SERBP1 and pTau was confirmed using an *in situ* proximity ligation assay - **Figure 7B**. Quantification of the PLA signal showed increased SERBP1-pTau interaction in AD compared to Ctr - **Figure 7C**. Western blot analysis showed that SERBP1 displays increased expression in AD brains in comparison to normal controls. Interestingly, SERBP1 appears to be often present as a dimer - **Figure 7D-E**. Overall, our data suggest that SERBP1 shows increased accumulation in AD brains and condenses in hyperphosphorylated Tau aggregates.

146 proteins present in Tau aggregates were also identified as SERBP1 interactors - **Figure 7F** and **Table S9**. GO analysis indicated that this group is strongly associated with ribosome biogenesis, translation, and splicing - **Figure 7F**. snoRNAs are also aberrantly located in Tau aggregates (77). Adding to SERBP1 participation in Tau aggregates, we determined that 66% of the snoRNAs enriched in Tau aggregates were bound by SERBP1 according to our study - **Table S9**. Ribosome biogenesis and translation are impaired in AD (78, 79). SERBP1 may help trap snoRNAs and factors implicated in ribosome biogenesis and translation in Tau aggregates, but further investigation is required.

PARP1 activity is increased in AD brains. Immunostaining of PARP1 and PAR indicates that PARylation is hyperactivated in AD neurons. Ultimately, this hyperactivation can trigger proteins to condense into toxic aggregates that contribute to neurodegeneration (80). We investigated PARP1-SERBP1 association in normal vs. AD brains. PARP1 exhibited a stronger presence in AD cortices compared to controls. Furthermore, PARP1 appeared to co-localize with SERBP1 in the cytoplasmic fraction of AD samples (white arrowhead) - **Figure 8A**. In control samples, PARP1 and SERBP1 also co-localized, albeit to a lesser extent compared to AD samples (confirmed by PCC analysis) - **Figure 8A-B**. Fluorescence profiles corroborate their association - **Figure 8C**. We performed a PLA assay to measure physical interaction between SERBP1 and PARP1. PLA results showed an increased signal for SERBP1/PARP1 association in AD brains in comparison to control brains. This observation was confirmed by the quantification of positive PLA area covered by such association - **Figure 8D**.

G3BP1, a well-established regulator of stress granule assembly, has been identified here as a SERBP1-associated protein - **Table S1**. In AD brains, G3BP1 is present in pathological stress granules (81, 82). Co-immunofluorescence (IF) experiments in control and AD brain sections revealed a stronger fluorescent signal for SERBP1 and G3BP1 in AD compared to controls, indicating higher levels of SERBP1 protein in stress granules in pathological condition - **Figure S6A**. Moreover, SERBP1 IF showed a high density of puncta with some large aggregates in AD samples. To quantify the level of SERBP1 in stress granules in control and AD samples, we measured the number of positive SERBP1 puncta in G3BP1 stress granules, observing a significant increment of SERBP1 in stress granules in AD cases. The total number of G3BP1 and SERBP1 puncta indicated greater accumulation in AD tissues compared to control - **Figure S6B**. SERBP1 and G3BP1 also show strong co-localization in glioblastoma cells and highly correlated expression in different scenarios - **Figure S6C, Table S5**.

Discussion

SERBP1's name relates to its function as a regulator of the gene SERPINE1, which encodes the plasminogen activator inhibitor 1 (PAI-1) (83), a protein implicated in senescence and migration. SERBP1 is localized mostly in the cytoplasm, but it is also present in the nucleus.

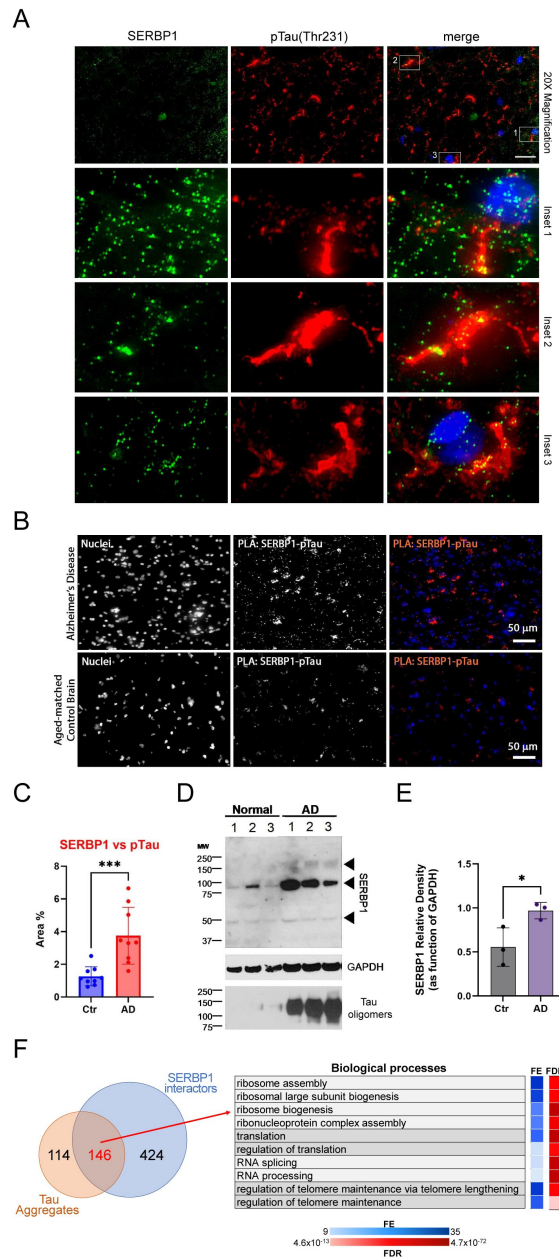
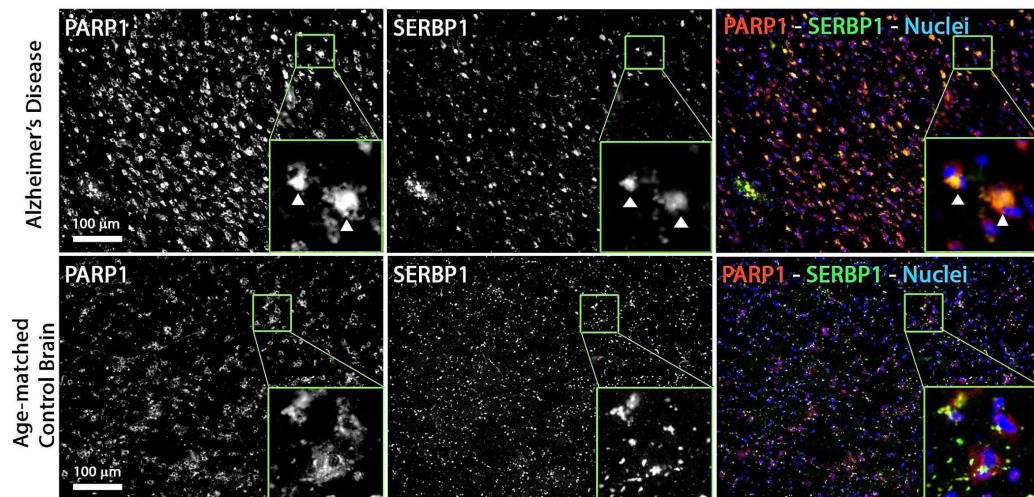


Figure 7

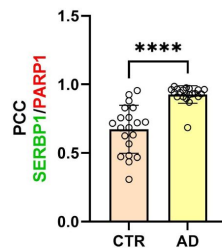
SERBP1 is present in Tau aggregates.

A) Representative co-immunofluorescence of SERBP1 and Phospho-Tau (Thr231) in AD brain tissues. Merged channel is represented. DAPI was used to stain nuclei. Magnification 20x and white scale bar: 50 μ m. Three different Insets selected from merged channels are represented in zoomed images as merged, SERBP1 (green) and Phospho-Tau (red). **B)** Representative PLA of SERBP1 vs. pTau in AD and aged-matched control brains (magnification 40x and white scale bar: 50 μ m). **C)** Percentage of positive area to PLA fluorescence in region of interests in AD and control brains (Ctrl vs AD, *** $p < 0.001$, paired t-test). **D)** Western blot showing SERBP1 expression in normal and AD brains and presence of oligomers. GAPDH and Tau oligomers immunoblot are shown. **E)** SERBP1 relative density in Ctr and AD brains, quantified as function of GAPDH (from immunoblot in D). (Ctrl vs. AD, * $p < 0.05$, paired t-test). **F)** Venn diagram shows overlap between proteins often identified in Tau aggregates and SERBP1 associated proteins identified in this study (top). Gene Ontology (GO) enrichment analysis (Biological Processes) using ShinyGO (27) indicated that SERBP1-associated factors that are also present in Tau aggregates are implicated in ribosome biogenesis, translation, splicing and telomere maintenance. FE = fold enrichment; FDR = false discovery rate. Datasets used to prepare the figure and detailed analyses are in Tables S1 and S9.

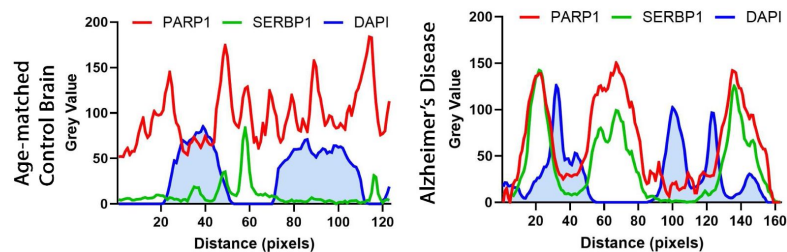
A



B



C



D

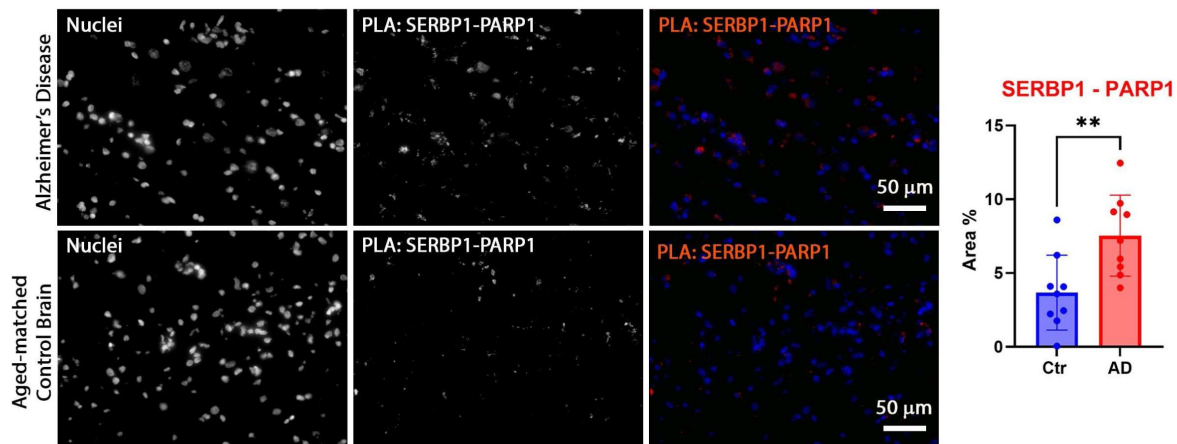


Figure 8

SERBP1 association with PARP1 in Alzheimer's brains.

A) Representative co-immunofluorescence of SERBP1 and PARP1 in AD and age-matched control brain tissues. PARP1 and SERBP1 are represented in gray while merged images represent PARP1 (red), SERBP1 (green). DAPI was used to stain nuclei (blue). Magnification 20x and white scale bar: 100 μ m. Each frame has a zoomed inset representing the detailed distribution of each target. **B)** Pearson Coefficient (PCC) of co-localizing SERBP1 and PARP1 between cells of age-matched control and AD brains (CTR vs. AD, **** $p < 0.001$ paired t-test). **C)** Fluorescence intensity profiles of PARP1 (red), SERBP1 (green), and DAPI (blue) in representative cells from age-matched control and AD brains. Distance is represented in pixels and fluorescence intensity as Grey value obtained using ImageJ Fiji software. **D)** Representative PLA of SERBP1 vs. PARP1 in AD and age-matched control brains (magnification 40x and white scale bar: 50 μ m). Percentage of positive area to PLA fluorescence in region of interests in AD and control brains (Ctr vs AD, ** $p < 0.01$, paired t-test).

SERBP1 cytoplasmic/nuclear balance changes during the cell cycle, and its nuclear presence intensifies upon stress (35, 37, 39). We have identified SERBP1's interactome and expanded its regulatory roles – particularly in the nucleus, where it is associated with specific complexes implicated in splicing, cell division, chromosome structure, and different aspects of ribosome biogenesis.

Most SERBP1-associated proteins are also RBPs. However, SERBP1 has unique characteristics; it is mostly disordered and does not display classic RNA binding domains other than RGG boxes (25). Two commonalities between SERBP1 and its associated proteins are their preference for G4 binding and the fact that they get PARylated and/or bind to PAR. These results indicate that complexes in which SERBP1 participates are assembled via G4 or PAR binding. Since SERBP1 does not display any apparent functional domains and is involved in very distinct regulatory complexes, we suggest that SERBP1 acts in the initial steps of their assembly via recognition of interacting sites in RNA, DNA, and proteins.

SERBP1 in translation regulation and ribosomal biogenesis

SERBP1 interacts with ribosomal proteins and regulates translation (33–36), functioning as a repressor or activator depending on the context. For instance, SERBP1 binding to CtIP mRNA during the S phase increases its translation (35), while changes in binding to the 40S ribosomal subunit during mitosis lead to PKCε pathway-mediated translation repression (36). Additionally, the SERBP1 yeast homolog, STM1, is required for optimal translation under nutritional stress

(84). SERBP1 migrates to stress granules under stress conditions (37). Recent findings indicated that translation in these organelles is not uncommon (85), suggesting the possibility of a similar role for SERBP1 as a modulator of translation in stress conditions.

Our GO analyses revealed that SERBP1 interacts with proteins participating in all stages of translation – initiation, elongation, and termination – including both negative and positive regulators. We found that 17% of SERBP1-associated proteins are implicated in translation initiation, including several subunits of the eukaryotic initiation factor 3 (eIF3A-I, L, and M) and CSDE1, among others. These data suggest that SERBP1's primary function in translation regulation lies in initiation. CSDE1 is a known regulator of both cap-dependent and cap-independent translation, functioning either as an enhancer (86, 87) or repressor (88, 89). The expression of CSDE1 is highly correlated with SERBP1 in multiple instances. CSDE1 was proposed to function as a “protein-RNA connector” enabling interactions between RNAs and regulators to control RNA's translational fate (90). A similar role could be ascribed to SERBP1, possibly explaining its different effects on translation.

Multiple SERBP1-associated proteins are involved in viral translation. For instance, the aforementioned interactors, components of the eIF3 complex and CSDE1, are both implicated in IRES-mediated viral translation in a variety of viruses (91–93). SERBP1 itself has also recently been shown to interact with dengue virus RNA and regulate its translation (94). Finally, we observed a significant overlap between SERBP1-associated proteins and the SARS-CoV-2 interactome, with 82 proteins out of 145 shared interactors being involved in translation.

SERBP1's interactors and its presence in the nucleolus suggest its participation in ribosome biogenesis. Besides binding to mRNA, SERBP1 binds to rRNA and snoRNAs, which play a role in pre-rRNA processing and modification (95). SERBP1, and 98 of its associated proteins, were previously identified in a screening for factors affecting rRNA processing (96). SERBP1's binding profile to rRNA and snoRNA is dynamic, as shown by differences in iCLIP sites in mitotic vs. asynchronous cells (36), hinting that SERBP1's participation in these processes is cell cycle-dependent as seen for translation. FBL, a SERBP1-associated protein, is involved in pre-rRNA processing, pre-rRNA methylation, and ribosome assembly (97). Recently, it has been shown that FBL also modulates rDNA transcription during the cell cycle according to its acetylation

status, leading to higher levels of rRNA synthesis in interphase and lower levels during mitosis (98 [↗](#)). SERBP1 also interacts with multiple proteins of the cohesin complex (SMC1, SMC3, and PDS5B) plus the associated factor, CTCF. Cohesin forms an enhancer boundary complex whose binding upstream of the 47S rDNA is as an early event in the activation of rDNA transcription (99 [↗](#)). In accordance with this finding, mutations in cohesin components have been linked to defects in both rRNA production and processing in yeast and human cells (100 [↗](#)). Two of SERBP1's top interactors, the RNA helicase DDX21 and NCL, also promote rDNA transcription (101 [↗](#), 102 [↗](#)).

Overall, these findings indicate a rather fluid role for SERBP1 in translation regulation, promoting or repressing different translational steps and acting in rRNA processing/modification and ribosome biogenesis through diverse pathways.

SERBP1 and PARylation

PARylation of SERBP1 was described in several different studies (62 [↗](#), 63 [↗](#), 103 [↗](#)–106 [↗](#)). Moreover, the expression of human PARP1 in *S. cerevisiae*, which does not naturally express PARP proteins, caused PARylation of many proteins implicated in ribosomal biogenesis, including STM1, SERBP1 homolog (107 [↗](#)). Our results suggest a two-way association between SERBP1 and PARP1. First, SERBP1 and 56% of its identified associated proteins either get PARylated and/or bind PAR (62 [↗](#)–64 [↗](#)). This result suggests that SERBP1 participates in several regulatory complexes modulated via PARylation.

PARylation of RBPs ultimately influences RNA processing and expression (59 [↗](#), 108 [↗](#)). In particular, PARylation of splicing regulators is known to affect alternative splicing. For instance, PARylation of HRP38's RNA binding domain decreases its binding activity and subsequently its ability to inhibit splicing (59 [↗](#)). PARylation also affects protein-protein interaction as observed in the case of hnRNPA1 (109 [↗](#)). Finally, PARP1 itself is involved in splicing. PARP1 binds RNA with a preference for intronic regions, while its knockdown affected many splicing events (110 [↗](#)). Several splicing regulators known to get PARylated and/or bind to PAR were identified as SERBP1-associated proteins, including hnRNPs (A1, H1, A2B1, H3, K, I, and U) and SR proteins (SRSF2, SRSF3, and SRSF9). In particular, most splicing alterations observed in SERBP1 KD cells were also present upon hnRNPU KD, suggesting that they work as co-factors in splicing regulation. Altogether, the data suggest that SERBP1 is directly implicated in splicing regulation and is present in complexes modulated by PARylation.

PAR induces phase separation of proteins containing IDRs and the formation of molecular condensates or membraneless organelles such as stress granules, mitotic spindles, DNA repair foci, and nucleoli (111 [↗](#)). Similarly, ADP-ribose can be conjugated to polar and charged amino acids, which are often found in IDRs (111 [↗](#)). Ultimately, PAR serves as a scaffold to assemble protein complexes in different contexts where PAR length and structure influence their composition (111 [↗](#)). The large number of PARylated proteins and PAR binders found among SERBP1-associated proteins strongly indicates that PAR is a critical component of the regulatory complex in which SERBP1 participates, especially in membraneless organelles. As a protein that interacts with PARP1, promotes PARP activity, gets PARylated, and binds PAR, we suggest that SERBP1 acts as a critical factor in the assembly of PARylation-dependent complexes. Additional analyses are required to assess whether the formation of such complexes depends on SERBP1 and if inhibition of SERBP1 PAR binding activity could be explored as a therapeutic strategy.

SERBP1 exhibited a punctate distribution in normal brain, indicating its presence as functional singlet condensate. However, in AD brain tissues, SERBP1 appeared to accumulate in large cytoplasmic aggregates and to strongly co-localize with PARP1. This result suggests that PARP1 modulates liquid-liquid phase separation (LLPS) and a PARylation-mediated mechanism is likely

involved in SERBP1's pathological transition from condensate to aggregate. Several RBPs containing intrinsically disordered regions, like SERBP1, have been shown to display alterations in LLPS in neurodegenerative diseases ([112](#), [113](#)).

SERBP1 and G4 binders

SERBP1 belongs to a group of neuronal-related RBPs that share structural characteristics and preferentially bind to G quadruplexes (G4s). G4s are complex DNA or RNA structures that display stacked tetrads of guanines stabilized by Hoogsteen base pairing. rG4s function as regulatory elements in different stages of gene expression, including splicing, mRNA decay, polyadenylation, and translation. G4s have been linked to cancer, viral replication, and in particular neurodegenerative diseases and neurological disorders ([114](#)–[116](#)).

The SERBP1 interactome shows an interesting parallel between PAR and G4 binding. Almost half of the SERBP1-associated proteins that bind G4 also bind PAR. PAR and G4s occupy the same regulatory space that includes DNA repair and stress response, and both induce phase separation. However, not much is known regarding the association between G-quadruplex and PAR binding/PARYlation, aside from the fact that PARP1 binds G4s, and this leads to the activation of its enzymatic activity ([117](#)). We envision two possible scenarios for SERBP1 and its associated proteins. Two types of domains have been implicated in both PAR and G4 binding, RGG boxes and RRM. Among 105 SERBP1-associated factors that bind PAR and G4, 51 contain RGG and/or RRM domains. Proteins that employ the same domain to bind PAR and G4 could transit from G4 to PAR binding complexes, depending on the cellular state. In a different setting, binding to G4 motifs in DNA or RNA could serve as a starting point for the assembly of PAR binding/PARYlated protein complexes. Further investigation on PAR- and G4-dependent interactions and identification of G4 and PAR binding domains in proteins present in these complexes are needed to elucidate the relationship between PAR and G4 binding.

SERBP1 potential role in neurodegenerative diseases and neurological disorders

SERBP1 expression in the brain is lower than in other organs and is increased in glioblastoma ([26](#)) and Alzheimer's brains, as shown in this study. SERBP1 is present in Tau aggregates ([9](#)) and in pathological stress granules in connection with G3BP1. PAR levels regulate the dynamics of RNPs and aberrant PARP1 activity promotes aggregation ([109](#)). Over 60% of SERBP1-associated proteins detected in Tau aggregates get PARYlated and/or bind PAR ([62](#)–[64](#)) and increased SERBP1-PARP1 interaction was observed in AD brains. Besides its potential role in aggregate formation, increased expression of SERBP1 in Alzheimer's brains could lead to important changes in gene expression. SERBP1 is highly expressed in neuronal and glioma stem cells and its levels drop significantly during neurogenesis. SERBP1 transgenic expression disrupted neuronal differentiation while its knockdown increased the expression of genes implicated in nervous system development, synaptic signaling, organization, memory, learning, and behavior ([26](#)).

SERBP1-associated proteins contain several factors involved in neurological disorders and neurodegenerative diseases, including FMR1, SMN1, and PABPN1. SERBP1 interaction with FMRP (FMR1) has been previously described ([36](#), [37](#)). Not only was this interaction validated, but we also determined that SERBP1 associates with FMRP partner proteins FXR1 and FXR2. FMRP is the main driver of Fragile X mental retardation but has also been linked to Alzheimer's disease and autism ([118](#), [119](#)). FMRP is a complex protein that binds different RNA motifs (including G4s) and regulates translation, splicing, RNA editing, and miRNA function, affecting synapse formation and plasticity, stress granule formation, channel signaling, and differentiation ([120](#)–[123](#)). Previous results indicate that SERBP1 and FMRP interact in different contexts ([36](#), [37](#)), suggesting that they function together in diverse stages of gene expression.

Spinal Muscular Atrophy (SMA) is caused by mutations in the SMN1 gene. Patients show progressive lower motor neuron loss (6). We observed associations between SERBP1, SMN1, and other members of the SMN complex, GEMIN4 and 5. SMN1 brings together the complex and is essential for spliceosome assembly. Since motor neurons are particularly sensitive to SMN1 mutations, additional roles for the SMN complex in these cells are proposed (6).

SYNCRIP and hnRNPU were identified among the top proteins associated with SERBP1 based on our proteomics analyses and gene expression correlations. SYNCRIP plays multiple roles in neuronal cells, including dendritic translation, synaptic plasticity, axonogenesis, and morphology (124–128). hnRNPU has been implicated in cortex development (129). Mutations in both hnRNPU and SYNCRIP are linked to intellectual disability (130–132) and neurodevelopmental disorders (133, 134), and were the two top candidates in an IBM Watson-based study to identify RBPs altered in amyotrophic lateral sclerosis (ALS) (135).

SERBP1 has not been studied in the context of the nervous system, and its contribution to neurodegenerative diseases and neurological disorders remains to be investigated. Our results suggest that SERBP1 is “guilty by association” as it interacts with several RBPs that function as key players in these scenarios. Moreover, SERBP1 regulates the expression of genes implicated in neuronal differentiation and synaptogenesis, processes often compromised in disease states. In the case of Alzheimer’s disease, increased expression of SERBP1 and its presence in pathological aggregates hint at a potential involvement. Additional studies are necessary to establish causal connections that link SERBP1 to AD phenotypes.

Material and methods

Cell culture, transfection, knockdown

U251 and U343 glioblastoma cell lines were obtained from Uppsala University (Uppsala, Sweden). 293T cells were obtained from the American Type Tissue Collection (ATCC). All cell lines were cultured in DMEM medium (HyClone, Cat# SH30243.01) supplemented with 10% FBS (Corning, Cat# 35015CV) and 1% penicillin/streptomycin (Gibco, Cat# 10378016). Cells were maintained at 37°C in a 5% CO₂ atmosphere. Cell lines were confirmed to be correct using Short Tandem Repeat profiling. They were regularly tested for mycoplasma contamination. Knockdown of SERBP1 (siSERBP1 KD) in U251 and U343 cells was performed by reverse transfection with 25 nM of control or SERBP1 SMARTpool siRNA (Dharmacon, Cat# L-020528-01-0005) using Lipofectamine RNAiMAX (Invitrogen, Cat# 13778150).

Plasmid constructs

A vector containing the Streptavidin-Binding Peptide (SBP)-Tag was prepared in pEF1 (Thermo Fisher, Cat# V 92120), and SERBP1 ORF was cloned in frame with the His-Myc-tag present in the vector to create pSBP-SERBP1. The SBP-tag contains 38 amino acids and binds to streptavidin with an equilibrium dissociation constant of 2.5 nM (136). Ultra-SERBP1 lentiviral vector (26) was used as a control in pulldown experiments. SERBP1 ORF was inserted into the pcDNA3.1-mGreenLantern plasmid (Addgene Plasmid #161912) to make mGreen-SERBP1. Flag-PARP1 was obtained from Addgene (#111575).

Primer name	sequence
SERBP1_for_EcoRI	tatagaattcATGCCTGGGCACTTAATGCCTGGGCACTTA
SERBP1_rev_XbaI	tatatctagaAGCCAGAGCTGGGA
mGL_SERBP1_BsrGI	actgtacaagATGCCTGGGCACTTACAG
mGL_SERBP1_XbaI	atctagattaAGCCAGAGCTGGGAA

Identification of SERBP1 interactors

293T cells were seeded (triplicate) in 10 cm dishes the day before transfection. pSBP-SERBP1 or pUltra-SERBP1(control) plasmids were transfected using CaCl_2 . 48 hours later, cells were collected and washed twice in cold PBS. Total protein extract was prepared by using TNE buffer (Tris pH 7.4 10 mM, NP-40 1%, NaCl 150 mM, and phosphatase inhibitor cocktails, Thermo Fisher Scientific, Cat #78430), incubating cells for 30 minutes on ice, and then sonicating five times for 3 seconds at 20% amplitude and 10-second intervals. A second batch of samples was prepared by using polysomal lysis buffer (KCl 100 mM, EDTA 25 mM, MgCl_2 5 mM, HEPES pH 7.0 10 mM, NP-40 0.5%, DTT 2 mM, VRC 0.4 mM, glycerol 10%, and phosphatase inhibitor cocktails, Thermo Fisher Scientific). Cell lysates were centrifuged at 4°C to eliminate debris and supernatants were collected and used later in pulldown assays.

50 μl of packed streptavidin beads (GE Healthcare Life Sciences, Cat# 17-5113-01) were washed two times in PBS and subsequently blocked in 1% BSA/PBS for 30 minutes at 4°C. Beads were finally washed five times in TNE buffer and then combined with cell extracts. The solution was incubated at room temperature for 2 hours and beads were later recovered via centrifugation at 4°C. The supernatant was discarded, and beads were washed five times with 1 ml of TNE buffer. To elute proteins, samples were resuspended in elution buffer (Tris pH 7.4 50 mM, NaCl 250 mM, NP-40 0.5%, deoxycholate 0.1%, 10 mM biotin), mixed, and incubated for 45 min at 37°C in a thermo-shaker. The supernatant was collected after centrifugation for mass spectrometry analysis.

Protein identification by mass spectrometry

Analysis of proteins associated with SERBP1

Proteins were separated by 1D SDS-PAGE using a Criterion XT 12% gel that was electrophoresed for ~1.5 cm and then stained with Coomassie blue (one experiment per gel). The protein-containing region of each gel lane was divided into six slices which were individually reduced *in situ* with TCEP [tris(2-carboxyethyl)phosphine] and alkylated in the dark with iodoacetamide prior to treatment with trypsin. Each digest was analyzed by capillary HPLC-electrospray ionization tandem mass spectrometry on a Thermo Scientific Orbitrap Fusion Lumos mass spectrometer. On-line HPLC separation was accomplished with an RSLC NANO HPLC system (Thermo Scientific/Dionex): column, PicoFrit™ (New Objective; 75 μm i.d.) packed to 15 cm with C18 adsorbent (Vydac; 218MS 5 μm , 300 Å). Precursor ions were acquired in the Orbitrap mass spectrometer in centroid mode at 120,000 resolution (m/z 200); data-dependent higher-energy C-trap dissociation (HCD) spectra were acquired at the same time in the linear trap using the “top speed” option (30% normalized collision energy). Mascot (v2.7.0; Matrix Science) was used to search the spectra against a combination of the human subset of the UniProt database plus a database of common contaminants [UniProt_Human 20181204 (95,936 sequences; 38,067,061 residues)]; contaminants 20120713 (247 sequences; 128,130 residues)]. Subset search of the identified proteins by X! Tandem, cross-correlation with the Mascot results, and determination of protein and peptide identity probabilities were accomplished by Scaffold (v4.9.0; Proteome Software). The thresholds for acceptance of peptide and protein assignments in Scaffold were set to yield < 1% protein FDR. Corresponding UniProt IDs for the proteins were obtained using the UniProt ID mapping service ([71](https://www.uniprot.org/mapping) ).

Western blots

Cells

Cells were harvested and lysed in Laemmli sample buffer (BioRad, Cat# 1610737). Cell extracts were separated on an SDS-PAGE gel, and transferred to PVDF membranes, previously activated with methanol. Membranes were blocked in 5% milk Tris-buffered saline with Tween 20 and probed with a collection of different antibodies listed below. Horseradish peroxidase (HRP)-conjugated goat anti-rabbit antibody (Santa Cruz Biotechnology, Cat# sc-2030) or HRP-conjugated goat anti-mouse (Santa Cruz Biotechnology, Cat# sc-2005) were used as secondary antibodies. Immobilon Western chemo-luminescence substrate (Millipore, Cat# WBKLS0500) was used to detect selected proteins.

Human samples

Western blots were performed with proteins extracted from control and Alzheimer's brains. Approximately 10 µg of protein were loaded into precast NuPAGE 4-12% Bis-Tris gels (Invitrogen) for analysis by sodium dodecyl sulfate–polyacrylamide gel electrophoresis. The separated proteins were subsequently transferred onto nitrocellulose membranes and blocked overnight at 4°C with 10% nonfat dry milk. The membranes were then probed for 1 hour at room temperature with primary antibodies - anti-SERBP1 (1:100, sc100800) and anti-glyceraldehyde 3-phosphate dehydrogenase (GAPDH) (1:1000, ab9485, Abcam) - diluted in 5% nonfat dry milk. SERBP1 immunoreactivity was detected with a horseradish peroxidase–conjugated anti-mouse IgG (1:5000, GE Healthcare). GAPDH immunoreactivity was detected using an anti-mouse IgG (1:6000, GE Healthcare) diluted in 5% nonfat dry milk. An enhanced chemiluminescence reagent (ECL Plus, Amersham) was used to visualize the bands. The amount of protein was normalized and quantified using the loading control GAPDH.

Antibodies

anti-SERBP1 (Santa Cruz Biotechnology, Cat# SC100800), anti-G3BP1 (Cell Signaling, Cat# 17798S), anti-PARP1 (Cell Signaling, Cat# 9532S), anti-SYNCRIP (Invitrogen, Cat# PA5-59501), anti-hnRNP (Cell Signaling, Cat# 34095), anti-Puromycin (Kerafast, Cat# EQ0001), anti-His (Santa Cruz Biotechnology, Cat# sc-53073), anti-Flag (Invitrogen, Cat# MA1-91878), anti-GAPDH (Santa Cruz Biotechnology, Cat# SC32233), anti-β-Actin (Abcam, Cat# ab8227), anti-β-Tubulin (Sigma-Aldrich, Cat# T8328), anti-α-Tubulin (Invitrogen, Cat# 236-10501), FBL (Cell Signaling, Cat# C13C3), NCL (Cell Signaling, Cat# D4C70).

High-throughput drug screening

High-throughput screening was conducted with the Cambridge Cancer Compound Library (Selleck Chem, Cat# L2300) of 293 compounds of interest (Table S7). All compounds dissolved in DMSO to a stock concentration of 10mM and further diluted in PBS to achieve a final treatment concentration of 100 nM and a final DMSO concentration of 0.1%. Four replicates of both U343 OE and U343 Control cells were plated in 384 well white polystyrene microplates (Thermo Scientific) at a seeding density of 5.0×10^3 cells/well 12 hours before treatment. Cell seeding and treatment was conducted by an automated electronic pipetting system to increase pipetting accuracy and to minimize variability between replicates (Integra VIAFLO 384). After 24 hours of treatment, cells were incubated with Cell Titer Glo 2.0 (Promega, Cat# G9243) according to the manufacturer's protocol. The addition of the reagent lyses cells and attaches a luminescent tag to ATP molecules in order to assess cell viability. Within 10 minutes of Cell Titer Glo 2.0 reagent addition, ATP measurements were then collected as luminescence intensities by a Tecan Spark microplate reader. Dixon's Q test was used to eliminate outliers prior to statistical analysis. Experimental ATP intensities were normalized to respective vehicle controls to assess cell viability following drug treatment. Student's T test was used to evaluate significant differences between reactions of U343 and U343 SERBP1 OE to each treatment.

Puromycin assays

siSERBP1 KD in U251 and U343 cells was performed as described above. 72 hours after transfection, cells were incubated with 10 $\mu\text{g/ml}$ Puromycin (Sigma-Aldrich, Cat# P7255) for 15 minutes at 37 °C and 5% CO₂. Cells were then placed on ice and washed twice with cold PBS before being harvested in Laemmli sample buffer. Western blots were performed as described above with 15 μg protein extracts.

Protein co-localization

U251 cells were cultured on glass coverslips until 80% confluent and fixed in 4% paraformaldehyde followed by permeabilization with 0.1% TritonX-100 for 2Lminutes. Cells were then stained with primary antibody for 1 hour followed by the secondary antibody conjugated to Alexa Fluor 488 or 568 (Invitrogen, Cat# A11004 and A11008). After washing with PBS, cells were stained with DAPI and mounted onto glass slides with FluorSave (Invitrogen, Cat# 345789) before being analyzed by fluorescence microscopy.

Impact of SERBP1 on cell division

24 hours after siSERBP1 KD in U251 and U343, cells were exposed to 20 nM of Paclitaxel (Cayman Chem, Cat# 10461) for 24 hours. Cells were fixed in 4% PFA and stained with α -Tubulin and DAPI. The experiment was performed in triplicates and 1000 cells were counted per replicate. Statistical significance was evaluated by Student's t-test.

In vitro PARylation assay

Flag-tagged PARP1 was expressed in insect cells and purified as previously described (137 [link](#)). 6xHis-tagged SERBP1 was purified as in (25 [link](#)). Briefly, purified recombinant PARP1 (0.1 μM) and SERBP1 (0.6 μM) were combined in an ADP-ribosylation reaction (20 μL) with or without sheared salmon sperm DNA (ssDNA) (100 ng/ μL) and NAD⁺ (100 μM). The reactions were incubated at room temperature for 15 minutes. The reaction products were analyzed on 8% PAGE-SDS gels with silver staining or Western blotting for PAR using a recombinant anti-poly-ADP-ribose binding reagent (Millipore, Cat# MABE1031; RRID: AB_2665467).

SERBP1 impact on PARylation

pcDNA3.1-EF1a-mGreenLantern-SERBP1 or control plasmid pcDNA3.1-EF1a-mGreenLantern was transfected into 293T cells. 48 hours later, cells were collected and lysed in Laemmli sample buffer. 10 μg of extracted proteins were separated on SDS-PAGE gel and Western blot was performed as described. Poly-ADP-ribose binding reagent (Sigma-Aldrich, Cat# MABE1031) was used to detect oligo- and poly-ADP-ribosylated (PARylated) proteins.

Sensitivity to PARP inhibitor

U251 and U343 siSERBP1 knockdown was performed as described above. 5 μM of PARP inhibitor PJ34 (Enzo, Cat# ALX-270-289) was used to treat the cells 24 hours later. The Essen Bioscience IncuCyte automated microscope system was used to follow the proliferation of U251 and U343 glioblastoma cells over 150 hours. Differences in proliferation between single and combined treatment at 100 or 150 hours were analyzed graphically. Statistical significance was calculated by Student's t-test and the combination treatment was evaluated with the Combination Index (138 [link](#)).

Immunolabeling of fixed human brain sections

Frozen cortical sections were first fixed in chilled 20% methanol. To eliminate autofluorescence from lipofuscin, we incubated the sections with TrueBlack® Lipofuscin Autofluorescence Quencher (#23007, Biotium) for 10 minutes at room temperature. After three washes with ethanol

70%, the sections were permeabilized in PBS/0.4% Triton-X100 for 5 minutes. After 1 hour of blocking in PBS/10% NGS/0.2% Triton at room temperature, the sections were incubated in primary antibodies diluted in PBS/10% NGS overnight at 4°C. Immunolabeling for G3BP1 (5 µg/ml, 2F3, Abcam Cat# ab56574), Phospho-Tau (Thr231) (1:500, AT180, Thermo Fisher Scientific #MN1040) and SERBP1 (1:50, 1B9, Santa Cruz Bio Cat# sc-100800) was performed in all control (N=3) and AD (N=3, Braak Stage V-VI) human brains. The next day, sections were washed three times in PBS (5 minutes each), and secondary antibodies were applied for 45 minutes: Alexa Fluor 488 and 568 (1:200, Life Technologies). After further washing in PBS, slides were then mounted with Prolong Gold Antifade with DAPI (Thermo Fisher Scientific Cat# P36931) to stain the nuclei. Granule density of G3BP1 and SERBP1 were measured in 3 Ctr and 3 AD cases, and images in triplicate from each case were analyzed by ImageJ FIJI (NIH) software. SERBP1 associated with stress granules was measured as the ratio between colocalizing SERBP1/G3BP1 puncta and G3BP1 total granules in the cortical section using ImageJ FIJI (NIH) software.

Human tissue harvesting

Frontal cortices of frozen brain tissues from age-matched control subjects (N=6), AD cases (N=6) were received as frozen blocks from the Institute for Brain Aging and Dementia at UC Irvine, approved by the Institutional Ethics Committee. Brain tissues were homogenized in 1X PBS mixed with a protease inhibitor cocktail (Roche) and phosphatase inhibitor (Sigma) at 1:3 (w/v) dilution of brain: PBS. Samples were then centrifuged at 10000 rpm for 20 min at 4°C. The supernatants, PBS-soluble fractions were aliquoted, snap-frozen, and stored at -80°C until use. The pellets were resuspended in the homogenization buffer (1X PBS) and were considered as insoluble fractions. They were also aliquoted and frozen at -80°C until use. Brain tissues analyzed in this study from diseased and age-matched non-demented control subjects are listed in Table 1.

Proximity ligation assay (PLA)

Detection of protein-protein interactions was conducted by an in-situ proximity ligation assay. The method depends on the recognition of target molecules in close proximity (<40 nm) by pairs of affinity probes, giving rise to an amplifiable detection signal. Briefly, PLA in mouse brain tissue has been performed using Duolink® PLA in Situ Red starter kit mouse/rabbit (Sigma Aldrich, DUO92101). The staining was performed following each protocol passage without modifications. Time of incubation and concentration of antibodies is established from IF protocol. Primary antibodies used for in situ proximity assay were: SERBP1 (1:100), PARP1 (1:200), and pThr231-Tau (1:500). Amplified red signal has been detected using Keyence Microscope.

Venn diagram statistics

Venn diagrams were created using online software (<https://bioinforx.com/apps/venn.php>). P-values for overlaps between two groups were calculated with a separate online tool (http://nemates.org/MA/progs/overlap_stats.html).

Gene ontology and network analyses

Gene Ontology (GO) enrichment analyses were performed using ShinyGO (versions 0.67 and 0.77), Metascape (version 3.5), and Panther (version 17.0) ([27](#), [28](#), [72](#)). Associations between GO terms were established using Revigo ([29](#)). Network analyses were conducted using STRING (version 11.5) and Metascape ([28](#), [139](#)) and visualized using Cytoscape (version 3.9.1) ([140](#)).

Expression correlation analyses

We performed expression correlation analyses using resources in R2 (<http://r2.amc.nl>) to identify genes with a strong positive correlation with SERBP1 ($R \geq 0.3$, p-value ≤ 0.05 , Pearson correlation) in the TCGA glioblastoma set (RNA-Seq samples), sarcoma (TCGA), bladder urothelial carcinoma (TCGA), neuroblastoma (SEQC), pancreatic adenocarcinoma (TCGA), and normal brain

(Hardy, $R \geq 0.5$ was selected in this case due to the high number of correlated genes). Additional analyses were conducted using the Ivy GAP glioblastoma dataset (141) following their default parameters ($R \geq 0.3$).

The Cortecon resource (142) was used to identify genes that share similarity with SERBP1 regarding their expression profile during *in vitro* cortex development. SERBP1 is part of cluster 5 which is linked to pluripotency. Genes that were in clusters related to pluripotency and whose overall profile resembles cluster 5 were selected for further analyses.

Identification of protein features and domain

Pfam accession numbers were obtained using the UniProt ID mapping service (71). R code was used to count the frequency of each accession number and create the consolidated data with fold enrichment and *fdr*-values. All accession numbers with a count above the average were analyzed with the dcGO Enrichment mining service (143).

Databases and datasets used for characterization of SERBP1-associated proteins

STM1 interactors were downloaded from BioGRID (version 4.4) (32) and human orthologs were obtained using the WORMHOLE website (144). Proximity-dependent biotinylation (Bio-ID) data for SERBP1 were taken from (30, 31). Proteins of the SARS-CoV-RNA interactome were derived from (57) and those of the PARP1 interactome from (66). RGG box-containing proteins were obtained from (50), and RRM motif-containing proteins from UniProt (71). G4 binding proteins were extracted from (46, 47), PAR binding proteins were obtained from (64), and PARylated proteins were identified by high-throughput studies described in (62, 63). The MobiDB database (version 5.0) (45) was used for identifying IDR-containing proteins. Data on the Tau aggregate interactome were derived from different studies compiled by (9).

Splicing analyses

To identify splicing alterations induced by SERBP1 and hnRNPU knockdown, adapter sequences were first trimmed from raw RNA-Seq reads of control and knockdown samples (26). RNA-Seq reads were then aligned against the human reference genome (version GRCh38) (145) and matching GENCODE transcriptome (v29) (146) using STAR (default parameters; version 2.7.7.a) (147). High-quality mapped reads ($q > 20$) from control and knockdown samples were processed using rMATS (default parameters; version v4.1.2) (148) to characterize splicing events. Events were classified into exon skipping (SE), mutually exclusive exons (MXE), intron retention (RI), alternative donor site (A5SS) or alternative acceptor site (A3SS). Splicing events were considered significant if their FDR-adjusted *p*-values were below 0.05 and absolute delta Percent Splice Inclusion (deltaPSI) above 0.1. Sashimi plots of splicing events were created using the *rmats2sashimiplot* tool (<https://github.com/Xinglab/rmats2sashimiplot>). To assess SERBP1 binding to the identified splicing alterations, we obtained processed iCLIP data from (36). The presence of iCLIP sites in the splicing alterations induced by SERBP1 knockdown was evaluated considering a window of 100 nt around splicing events using the *bedtools intersect* software (149).

Identification of snoRNAs and scaRNAs bound by SERBP1

RIP-sequencing results for SERBP1 were obtained from (26). Sequencing reads were processed using Kallisto (default parameters, version 0.46.1) (150) with an index of 31 k-mers and GENCODE (v28) as the reference for the human transcriptome (146). Transcript abundance estimates were then collapsed into gene-level counts using the R package *tximport* (151). Differential gene expression analyses were performed using DESeq2 (152). We compared SBP-SERBP1 versus SBP-GST samples and used $\log_2$ fold change ≥ 0.5 and adjusted (false discovery rate, FDR) *p*-value < 0.05 to identify snoRNAs and scaRNAs preferentially associated with SERBP1.

Supporting information

[Supplementary Material](#)

[Supplementary Table 1](#)

[Supplementary Table 2](#)

[Supplementary Table 3](#)

[Supplementary Table 4](#)

[Supplementary Table 5](#)

[Supplementary Table 6](#)

[Supplementary Table 7](#)

[Supplementary Table 8](#)

[Supplementary Table 9](#)

[Supplementary Figure 1](#)

[Supplementary Figure 2](#)

[Supplementary Figure 3](#)

[Supplementary Figure 4](#)

[Supplementary Figure 5](#)

[Supplementary Figure 6](#)

[List of key reagents](#)

Data availability

The mass spectrometry datasets generated and analyzed during this study are available in the MassIVE repository as a partner with ProteomeXchange and can be accessed by:

1. MassIVE dataset identifier MSV000091749, <ftp://massive.ucsd.edu/v05/MSV000091749/>, ProteomeXchange dataset identifier PXD041664
2. MassIVE dataset identifier MSV000093355, <ftp://massive.ucsd.edu/v06/MSV000093355/>, ProteomeXchange dataset identifier PXD046851

RNAseq datasets are available at the European Nucleotide Archive (ENA) with the accession number PRJEB69681.

The following datasets were generated:

HOSPITAL SIRIO LIBANES. Raw sequencing data for RNAseq study in U251 cells (control vs. SERBP1 KD and control vs. hnRNP KD). PRJEB69681. <https://www.ebi.ac.uk/ena/browser/view/PRJEB69681>

Susan Weintraub. MassIVE MSV000091749 and MassIVE MSV000093355. Raw Mass spectrometry data for pulldown study (control vs. SERBP1) in 293T cells to identify proteins associated with SERBP1. <https://massive.ucsd.edu/ProteoSAFe/dataset.jsp?task=a20a200385114896be5854646c2149d>
<https://massive.ucsd.edu/ProteoSAFe/dataset.jsp?task=a775aabf74874af784b475db3a02c609>

Code to perform the analysis of SERBP1 impact on splicing can be found in the Open Science Framework: DOI 10.17605/OSF.IO/4EHZS

The following previously published data sets were used:

Youn, J. Y., Dunham, W. H., Hong, S. J., Knight, J. D. R., Bashkurov, M., Chen, G. I., Bagci, H., Rathod, B., MacLeod, G., Eng, S. W. M., Angers, S., Morris, Q., Fabian, M., Côté, J. F., & Gingras, A. C. (2018). High-Density Proximity Mapping Reveals the Subcellular Organization of mRNA-Associated Granules and Bodies; High-Confidence BioID Proximity Interactome for the Organization of RNA Biology. IDs (three sets) MSV00008141, MSV000081412, MSV000081413. In the public domain at MassIVE Repository <https://massive.ucsd.edu/>. <https://massive.ucsd.edu/ProteoSAFe/dataset.jsp?task=3d92815a96454ea2a13e7f147ec2b09e> <https://massive.ucsd.edu/ProteoSAFe/dataset.jsp?task=e774f511588c4e109aba6d76a6ee264a> <https://massive.ucsd.edu/ProteoSAFe/dataset.jsp?task=db890fe0ae804b0592d162a07894f2f0>.

Go, C. D., Knight, J. D. R., Rajasekharan, A., Rathod, B., Hesketh, G. G., Abe, K. T., Youn, J. Y., Samavarchi-Tehrani, P., Zhang, H., Zhu, L. Y., Popiel, E., Lambert, J. P., Coyaude, É., Cheung, S. W. T., Rajendran, D., Wong, C. J., Antonicka, H., Pelletier, L., Palazzo, A. F., Shoubbridge, E. A., ... Gingras, A. C. (2021). A proximity-dependent biotinylation map of a human cell; Mass spectrometry samples and results for the core BioID dataset. IDs (two sets) MSV000084359, MSV000084360. In the public domain at MassIVE Repository <https://massive.ucsd.edu/>. <https://massive.ucsd.edu/ProteoSAFe/dataset.jsp?task=2749913f600246d5907d344007def7ae>, <https://massive.ucsd.edu/ProteoSAFe/dataset.jsp?task=bd7ca8514fee41da806cf16babec35b>.

HOSPITAL SIRIO LIBANES. RNA sequencing of U251 SERBP1 knockdown cells and RIP sequencing for the identification of SERBP1 targets in 293T cells. PRJEB35774. <https://www.ebi.ac.uk/ena/browser/view/PRJEB35774>

Lee, S., Lee, Y. S., Choi, Y., Son, A., Park, Y., Lee, K. M., Kim, J., Kim, J. S., & Kim, V. N. (2021). Profile of the SARS-CoV-2 RNA interactome. PXD024808. <https://proteomecentral.proteomexchange.org/cgi/GetDataset?ID=PX024808>.

Mosler, T., Baymaz, H. I., Gräf, J. F., Mikicic, I., Blattner, G., Bartlett, E., Ostermaier, M., Piccinno, R., Yang, J., Voigt, A., Gatti, M., Pellegrino, S., Altmeyer, M., Luck, K., Ahel, I., Roukos, V., & Beli, P. (2022). PARP1 proximity proteomics reveals interaction partners at stressed replication forks. PXD037154. <https://proteomecentral.proteomexchange.org/cgi/GetDataset?ID=PX037154>.

Herviou, P., Le Bras, M., Dumas, L., Hieblot, C., Gilhodes, J., Cioci, G., Hugnot, J. P., Amedan, A., Guillonneau, F., Dassi, E., Cammas, A., & Millevoi, S. (2020). hnRNP H/F drive

RNA G-quadruplex-mediated translation linked to genomic instability and therapy resistance in glioblastoma. PXD015609. <https://www.ebi.ac.uk/pride/archive?keyword=PX015609>.

Jernej Ule, Peter Parker. SERBP1 iCLIP in mitotic and asynchronous cells with Blu577 treatment. E-MTAB-10830. <https://www.ebi.ac.uk/biostudies/arrayexpress/studies/E-MTAB-10830>

Funding

This work was supported by the Cancer Prevention and Research Institute of Texas [RP200595 to LP], Alzheimer's Association [AARGD-NTF-22-968603 to LP] and a Mays Cancer Center Pilot Grant [to LP]. PAFG was supported by grants from Serrapilheira Foundation and Fundação de Amparo à Pesquisa do Estado de São Paulo (FAPESP) [2018/15579-8]. MM was sponsored by the Alzheimer's Association [AARF-21-720991]. GDAG was supported by a fellowship from FAPESP [2017/19541-2] and Young Scientist program, Hospital Sírio-Libanês. KB and ML were sponsored by the The German Academic Exchange Service (DAAD). AK was supported by NIH Supplement 2R01 HG006015 and the Greehey Foundation. AB was partially supported by a Voelker Fund Young Investigator Grant to DSL.

Conflict of interest

The authors declare no competing interests.

Acknowledgements

Mass spectrometry analyses were conducted at the Institutional Mass Spectrometry Laboratory, University of Texas Health Science Center, supported in part by NIH grant P30 CA54174-23 (S.T. Weintraub, Mays Cancer Center Mass Spectrometry Shared Resource) and the University of Texas System Proteomics Core Network for purchase of the Orbitrap Fusion Lumos mass spectrometer, with expert technical assistance of Sammy Pardo and Dana Molleur. We would like to thank Dr. Jernej Ule and Dr. Rupert Faraway for sharing the SERBP1 CLIP data and their valuable input. The authors would like to acknowledge Karen Klein for excellent editorial service.

Clinical Diagnosis	Case Number	Age	Gender	PMI (Hours)	Braak Tangles (0-6)	Application
AD	1154	86	M	3.25	6	IF/PLA
AD	1098	81	F	2.75	5	IF/PLA
AD	5773	74	M	10	5	IF/PLA
AD	5779	73	M	15	6	WB
AD	5781	88	M	10	5	WB
AD	5829	68	M	12	6	WB
Control	5263	88	M	12.17	1	IF/PLA
Control	1161	84	F	2.50	0	IF/PLA
Control	1106	79	M	1.75	2	IF/PLA
Control	1796	81	M	8	0	WB
Control	2-99	74	F	2.8	2	WB
Control	13-01	95	M	3.7	1	WB

IF: Immunofluorescence
 PLA: Proximity Ligation Assay
 PMI: Postmortem Interval
 WB: Western Blotting

Table 1: Brain tissues analyzed in this study from diseased and age-matched non-demented control subjects are summarized.

References

1. Gerstberger S., Hafner M., Tuschl T (2014) **A census of human RNA-binding proteins** *Nat Rev Genet* **15**:829–845 <https://doi.org/10.1038/nrg3813>
2. Corley M., Burns M.C., Yeo G.W (2020) **How RNA-binding proteins interact with RNA: molecules and mechanisms** *Mol Cell* **78**:9–29 <https://doi.org/10.1016/j.molcel.2020.03.011>
3. Kelaini S., Chan C., Cornelius V.A., Margariti A (2021) **RNA-binding proteins hold key roles in function, dysfunction, and disease** *Biology (Basel)* **10** <https://doi.org/10.3390/biology10050366>
4. Chan J.N., Sánchez-Vidaña D.I., Anoopkumar-Dukie S., Li Y., Benson Wui-Man L (2022) **RNA-binding protein signaling in adult neurogenesis** *Front Cell Dev Biol* **10** <https://doi.org/10.3389/fcell.2022.982549>
5. Bryant C.D., Yazdani N (2016) **RNA-binding proteins, neural development and the addictions** *Genes Brain Behav* **15**:169–186 <https://doi.org/10.1111/gbb.12273>
6. Prashad S., Gopal P.P (2021) **RNA-binding proteins in neurological development and disease** *RNA Biol* **18**:972–987 <https://doi.org/10.1080/15476286.2020.1809186>
7. Marcelino Meliso F., Hubert C.G., Favoretto Galante P.A., Penalva L.O. (2017) **RNA processing as an alternative route to attack glioblastoma** *Hum Genet* **136**:1129–1141 <https://doi.org/10.1007/s00439-017-1819-2>
8. Sanya D.R.A., Cava C., Onésime D (2023) **Roles of RNA-binding proteins in neurological disorders, COVID-19, and cancer** *Hum Cell* **36**:493–514 <https://doi.org/10.1007/s13577-022-00843-w>
9. Kavanagh T., Halder A., Drummond E (2022) **Tau interactome and RNA binding proteins in neurodegenerative diseases** *Mol Neurodegener* **17** <https://doi.org/10.1186/s13024-022-00572-6>
10. Velasco MX, Kosti A, Penalva LOF, Hernández G (2019) **The diverse roles of RNA-binding proteins in glioma development** *Adv Exp Med Biol* **1157**:29–39
11. Gebauer F., Schwarzl T., Valcárcel J., Hentze M.W (2021) **RNA-binding proteins in human genetic disease** *Nat Rev Genet* **22**:185–198 <https://doi.org/10.1038/s41576-020-00302-y>
12. Nussbacher J.K., Batra R., Lagier-Tourenne C., Yeo G.W (2015) **RNA-binding proteins in neurodegeneration: seq and you shall receive** *Trends Neurosci* **38**:226–236 <https://doi.org/10.1016/j.tins.2015.02.003>
13. An H., de Meritens C.R., Shelkova T.A. (2021) **Connecting the “dots”: RNP granule network in health and disease** *Biochim Biophys Acta Mol Cell Res* **1868** <https://doi.org/10.1016/j.bbamcr.2021.119058>

14. Drino A., Schaefer M.R (2018) **RNAs, phase separation, and membrane-less organelles: are post-transcriptional modifications modulating organelle dynamics?** *Bioessays* **40** <https://doi.org/10.1002/bies.201800085>
15. Gomes E., Shorter J (2019) **The molecular language of membraneless organelles** *J Biol Chem* **294**:7115–7127 <https://doi.org/10.1074/jbc.TM118.001192>
16. Zeke A., Schád É., Horváth T., Abukhairan R., Szabó B., Tantos A (2022) **Deep structural insights into RNA-binding disordered protein regions** *Wiley Interdiscip Rev RNA* **13** <https://doi.org/10.1002/wrna.1714>
17. Roden C., Gladfelter A.S (2021) **RNA contributions to the form and function of biomolecular condensates** *Nat Rev Mol Cell Biol* **22**:183–195 <https://doi.org/10.1038/s41580-020-0264-6>
18. Cave J.W., Willis D.E (2022) **33905176 G-quadruplex regulation of neural gene expression** *Febs J* **289**:3284–3303 <https://doi.org/10.1111/febs.15900>
19. Masai H., Tanaka T (2020) **G-quadruplex DNA and RNA: their roles in regulation of DNA replication and other biological functions** *Biochem Biophys Res Commun* **531**:25–38 <https://doi.org/10.1016/j.bbrc.2020.05.132>
20. Nakanishi C., Seimiya H (2020) **G-quadruplex in cancer biology and drug discovery** *Biochem Biophys Res Commun* **531**:45–50 <https://doi.org/10.1016/j.bbrc.2020.03.178>
21. Kosiol N., Juranek S., Brossart P., Heine A., Paeschke K (2021) **G-quadruplexes: a promising target for cancer therapy** *Mol Cancer* **20** <https://doi.org/10.1186/s12943-021-01328-4>
22. Darling A.L., Shorter J (2021) **Combating deleterious phase transitions in neurodegenerative disease** *Biochim Biophys Acta Mol Cell Res* **1868** <https://doi.org/10.1016/j.bbamcr.2021.118984>
23. Webber C.J., Lei S., Wolozin B. (2020) **The pathophysiology of neurodegenerative disease: disturbing the balance between phase separation and irreversible aggregation** *Prog Mol Biol Transl Sci* **174**:187–223 <https://doi.org/10.1016/bs.pmbts.2020.04.021>
24. Uversky V.N (2017) **The roles of intrinsic disorder-based liquid-liquid phase transitions in the “Dr. Jekyll–Mr. Hyde” behavior of proteins involved in amyotrophic lateral sclerosis and frontotemporal lobar degeneration** *Autophagy* **13**:2115–2162 <https://doi.org/10.1080/15548627.2017.1384889>
25. Baudin A., Moreno-Romero A.K., Xu X., Selig E.E., Penalva L.O.F., Libich D.S (2021) **Structural characterization of the RNA-binding protein SERBP1 reveals intrinsic disorder and atypical RNA binding modes** *Front Mol Biosci* **8** <https://doi.org/10.3389/fmolb.2021.744707>
26. Kosti A. et al. (2020) **The RNA-binding protein SERBP1 functions as a novel oncogenic factor in glioblastoma by bridging cancer metabolism and epigenetic regulation** *Genome Biol* **21** <https://doi.org/10.1186/s13059-020-02115-y>
27. Xijin Ge S., Jung D., Yao R. (2020) **ShinyGO: a graphical gene-set enrichment tool for animals and plants** *Bioinformatics* **36**:2628–2629 <https://doi.org/10.1093/bioinformatics/btz931>
28. Zhou Y., Zhou B., Pache L., Chang M., Khodabakhshi A.H., Tanaseichuk O., Benner C., Chanda S.K. (2019) **Metascape provides a biologist-oriented resource for the analysis of systems-level datasets** *Nat Commun* **10** <https://doi.org/10.1038/s41467-019-09234-6>

29. Supek F., Bošnjak M., Škunca N., Šmuc T (2011) **Revigo summarizes and visualizes long lists of gene ontology terms** *PLoS One* **6** <https://doi.org/10.1371/journal.pone.0021800>
30. Go C.D. *et al.* (2021) **A proximity-dependent biotinylation map of a human cell** *Nature* **595**:120–124 <https://doi.org/10.1038/s41586-021-03592-2>
31. Youn J.Y. *et al.* (2018) **High-density proximity mapping reveals the subcellular organization of mRNA-associated granules and bodies** *Mol Cell* **69**:517–532 <https://doi.org/10.1016/j.molcel.2017.12.020>
32. Stark C., Breitkreutz B.J., Reguly T., Boucher L., Breitkreutz A., Tyers M. (2006) **BioGRID: a general repository for interaction datasets** *Nucleic Acids Res* **34**:D535–D539 <https://doi.org/10.1093/nar/gkj109>
33. Brown A., Baird M.R., Cj Yip M., Murray J., Shao S (2018) **Structures of translationally inactive mammalian ribosomes** *Elife* **7** <https://doi.org/10.7554/eLife.40486>
34. Muto A., Sugihara Y., Shibakawa M., Oshima K., Matsuda T., Nadano D (2018) **The mRNA-binding protein Serbp1 as an auxiliary protein associated with mammalian cytoplasmic ribosomes** *Cell Biochem Funct* **36**:312–322 <https://doi.org/10.1002/cbf.3350>
35. Ahn J.W. *et al.* (2015) **SERBP1 affects homologous recombination-mediated DNA repair by regulation of CtIP translation during S phase** *Nucleic Acids Res* **43**:6321–6333 <https://doi.org/10.1093/nar/gkv592>
36. Martini S. *et al.* (2021) **A genetically-encoded crosslinker screen identifies SERBP1 as a PKCε substrate influencing translation and cell division** *Nat Commun* **12** <https://doi.org/10.1038/s41467-021-27189-5>
37. Lee Y.J., Wei H.M., Chen L.Y., Li C. (2014) **Localization of SERBP1 in stress granules and nucleoli** *Febs J* **281**:352–364 <https://doi.org/10.1111/febs.12606>
38. Lee Y.J., Hsieh W.Y., Chen L.Y., Li C (2012) **Protein arginine methylation of SERBP1 by protein arginine methyltransferase 1 affects cytoplasmic/nuclear distribution** *J Cell Biochem* **113**:2721–2728 <https://doi.org/10.1002/jcb.24151>
39. Passos D.O., Bressan G.C., Nery F.C., Kobarg J (2006) **Ki-1/57 interacts with PRMT1 and is a substrate for arginine methylation** *Febs J* **273**:3946–3961 <https://doi.org/10.1111/j.1742-4658.2006.05399.x>
40. Coppolecchia R., Buser P., Stotz A., Linder P (1993) **A new yeast translation initiation factor suppresses a mutation in the eIF-4A RNA helicase** *EMBO J* **12**:4005–4011
41. Bleckmann A. *et al.* (2023) **Cytosolic RGG RNA-binding proteins are temperature sensitive flowering time regulators in Arabidopsis** *Biol Chem* **404**:1069–1084 <https://doi.org/10.1515/hsz-2023-0171>
42. Sonnhammer E.L., Eddy S.R., Durbin R (1997) **Pfam: a comprehensive database of protein domain families based on seed alignments** *Proteins* **28**:405–420 [https://doi.org/10.1002/\(SICI\)1097-0134\(199707\)28:3<405::AID-PROT10>3.0.CO;2-L](https://doi.org/10.1002/(SICI)1097-0134(199707)28:3<405::AID-PROT10>3.0.CO;2-L)

43. Fonin A. V., Antifeeva I.A., Kuznetsova I.M., Turoverov K.K., Zaslavsky B.Y., Kulkarni P., Uversky V.N (2022) **Biological soft matter: intrinsically disordered proteins in liquid-liquid phase separation and biomolecular condensates** *Essays Biochem* **66**:831–847 <https://doi.org/10.1042/EBC20220052>
44. Su Q., Mehta S., Zhang J (2021) **Liquid-liquid phase separation: orchestrating cell signaling through time and space** *Mol Cell* **81**:4137–4146 <https://doi.org/10.1016/j.molcel.2021.09.010>
45. Piovesan D., Del Conte A., Clementel D., Monzon A.M., Bevilacqua M., Aspromonte M.C., Iserte J.A., Orti F.E., Marino-Buslje C., Tosatto S.C.E (2023) **MobiDB: 10 years of intrinsically disordered proteins** *Nucleic Acids Res* **51**:D438–D444 <https://doi.org/10.1093/nar/gkac1065>
46. Su H. *et al.* (2021) **Photoactive G-quadruplex ligand identifies multiple G-quadruplex-related proteins with extensive sequence tolerance in the cellular environment** *J Am Chem Soc* **143**:1917–1923 <https://doi.org/10.1021/jacs.0c10792>
47. Herviou P. *et al.* (2020) **hnRNP H/F drive RNA G-quadruplex-mediated translation linked to genomic instability and therapy resistance in glioblastoma** *Nat Commun* **11** <https://doi.org/10.1038/s41467-020-16168-x>
48. Huang Z.L., Dai J., Luo W.H., Wang X.G., Tan J.H., Chen S. Bin, Huang Z.S. (2018) **Identification of G-quadruplex-binding protein from the exploration of RGG motif/G-quadruplex interactions** *J Am Chem Soc* **140**:17945–17955 <https://doi.org/10.1021/jacs.8b09329>
49. Patra S., Claude J.B., Naubron J.V., Wenger J (2021) **Fast interaction dynamics of G-quadruplex and RGG-rich peptides unveiled in zero-mode waveguides** *Nucleic Acids Res* **49**:12348–12357 <https://doi.org/10.1093/nar/gkab1002>
50. Thandapani P., O'Connor T.R., Bailey T.L., Richard S (2013) **Defining the RGG/RG Motif** *Mol Cell* **50**:613–623 <https://doi.org/10.1016/j.molcel.2013.05.021>
51. Herdy B., Mayer C., Varshney D., Marsico G., Murat P., Taylor C., D'Santos C., Tannahill D., Balasubramanian S (2018) **Analysis of NRAS RNA G-quadruplex binding proteins reveals DDX3X as a novel interactor of cellular G-quadruplex containing transcripts** *Nucleic Acids Res* **46**:11592–11604 <https://doi.org/10.1093/nar/gky861>
52. McRae E.K.S., Dupas S.J., Booy E.P., Piragasam R.S., Fahlman R.P., McKenna S.A (2020) **An RNA guanine quadruplex regulated pathway to TRAIL-sensitization by DDX21** *RNA* **26**:44–57 <https://doi.org/10.1261/rna.072199.119>
53. Murat P., Marsico G., Herdy B., Ghanbarian A.T., Portella G., Balasubramanian S (2018) **RNA G-quadruplexes at upstream open reading frames cause DHX36- and DHX9-dependent translation of human mRNAs** *Genome Biol* **19** <https://doi.org/10.1186/s13059-018-1602-2>
54. Varshney D., Cuesta S.M., Herdy B., Abdullah U.B., Tannahill D., Balasubramanian S (2021) **RNA G-quadruplex structures control ribosomal protein production** *Sci Rep* **11** <https://doi.org/10.1038/s41598-021-01847-6>
55. Ruggiero E., Richter S.N (2018) **G-quadruplexes and G-quadruplex ligands: Targets and tools in antiviral therapy** *Nucleic Acids Res* **46**:3270–3283 <https://doi.org/10.1093/nar/gky187>
56. Xu J., Huang H., Zhou X (2021) **G-Quadruplexes in neurobiology and virology: functional roles and potential therapeutic approaches** *JACS Au* **1**:2146–2161 <https://doi.org/10.1021/jacsau.1c00451>

57. Lee S., Lee Y.S., Choi Y., Son A., Park Y., Lee K.M., Kim J., Kim J.S., Kim V.N (2021) **The SARS-CoV-2 RNA interactome** *Mol Cell* **81**:2838–2850 <https://doi.org/10.1016/j.molcel.2021.04.022>
58. Chung K.S., Won M., Lee J.J., Ahn J., Hoe K.L., Kim D.U., Song K. Bin, Yoo H.S. (2007) **17346842 Yeast-based screening to identify modulators of G-protein signaling using uncontrolled cell division cycle by overexpression of Stm1** *J Biotechnol* **129**:547–554 <https://doi.org/10.1016/j.jbiotec.2007.01.007>
59. Kim D.S., Challa S., Jones A., Kraus W.L (2020) **PARPs and ADP-ribosylation in RNA biology: from RNA expression and processing to protein translation and proteostasis** *Genes Dev* **34**:302–320 <https://doi.org/10.1101/gad.334433.119>
60. Gupte R., Liu Z., Kraus W.L (2017) **PARPs and ADP-ribosylation: recent advances linking molecular functions to biological outcomes** *Genes Dev* **31**:101–126 <https://doi.org/10.1101/gad.291518>
61. Huang D., Kraus W.L (2022) **The expanding universe of PARP1-mediated molecular and therapeutic mechanisms** *Mol Cell* **82**:2315–2334 <https://doi.org/10.1016/j.molcel.2022.02.021>
62. Gibson B.A., Zhang Y., Jiang H., Hussey K.M., Shrimp J.H., Lin H., Schwede F., Yu Y., Kraus W.L (2016) **Chemical genetic discovery of PARP targets reveals a role for PARP-1 in transcription elongation** *Science* **353**:45–50 <https://doi.org/10.1126/science.aaf7865>
63. Martello R., Leutert M., Jungmichel S., Bilan V., Larsen S.C., Young C., Hottiger M.O., Nielsen M.L (2016) **Proteome-wide identification of the endogenous ADP-ribosylome of mammalian cells and tissue** *Nat Commun* **7** <https://doi.org/10.1038/ncomms12917>
64. Dasovich M., Beckett M.Q., Bailey S., Ong S.E., Greenberg M.M., Leung A.K.L (2021) **Identifying poly(ADP-ribose)-binding proteins with photoaffinity-based proteomics** *J Am Chem Soc* **143**:3037–3042 <https://doi.org/10.1021/jacs.0c12246>
65. Zhang C. *et al.* (2023) **A ribosomal gene panel predicting a novel synthetic lethality in non-BRCAness tumors** *Signal Transduct Target Ther* **8** <https://doi.org/10.1038/s41392-023-01401-y>
66. Mosler T. *et al.* (2022) **PARP1 proximity proteomics reveals interaction partners at stressed replication forks** *Nucleic Acids Res* **50**:11600–11618 <https://doi.org/10.1093/nar/gkac948>
67. Kim D.S., Camacho C. V., Nagari A., Malladi V.S., Challa S., Kraus W.L (2019) **Activation of PARP-1 by snoRNAs controls ribosome biogenesis and cell growth via the RNA helicase DDX21** *Mol Cell* **75**:1270–1285 <https://doi.org/10.1016/j.molcel.2019.06.020>
68. Mekishvili M., Matveeva E., Fondufe-Mittendorf Y (2017) **Methodology to identify poly-ADP-ribose polymerase 1 (PARP1)-mRNA targets by PAR-CLiP** *Methods Mol Biol* **1608**:211–228 https://doi.org/10.1007/978-1-4939-6993-7_15
69. Wei H., Yu X (2016) **Functions of PARYlation in DNA damage repair pathways** *Genomics Proteomics Bioinformatics* **14**:131–139 <https://doi.org/10.1016/j.gpb.2016.05.001>
70. Žaja R., Mikoč A., Barkauskaite E., Ahel I. (2012) **Molecular insights into poly(ADP-ribose) recognition and processing** *Biomolecules* **3**:1–17 <https://doi.org/10.3390/biom3010001>
71. Consortium UniProt (2023) **UniProt: The universal protein knowledgebase in 2023** *Nucleic Acids Res* **51**:D523–D531 <https://doi.org/10.1093/nar/gkac1052>

72. Thomas P.D., Ebert D., Muruganujan A., Mushayahama T., Albou L.P., Mi H (2022) **PANTHER: Making genome-scale phylogenetics accessible to all** *Protein Sci* **31**:8–22 <https://doi.org/10.1002/pro.4218>
73. Li W., Jiang C., Zhang E (2021) **Advances in the phase separation-organized membraneless organelles in cells: a narrative review** *Transl Cancer Res* **10**:4929–4946 <https://doi.org/10.21037/tcr-21-1111>
74. Kastano K., Mier P., Dosztányi Z., Promponas V.J., Andrade-Navarro M.A (2022) **Functional tuning of intrinsically disordered regions in human proteins by composition bias** *Biomolecules* **12** <https://doi.org/10.3390/biom12101486>
75. Feng Z., Jia B., Zhang M (2021) **Liquid-liquid phase separation in biology: specific stoichiometric molecular interactions vs promiscuous interactions mediated by disordered sequences** *Biochemistry* **60**:2397–2406 <https://doi.org/10.1021/acs.biochem.1c00376>
76. Kimura T., Sharma G., Ishiguro K., Hisanaga S.I (2018) **Phospho-Tau bar code: analysis of phosphoisoforms of Tau and its application to tauopathy** *Front Neurosci* **12** <https://doi.org/10.3389/fnins.2018.00044>
77. Lester E., Ooi F.K., Bakkar N., Ayers J., Woerman A.L., Wheeler J., Bowser R., Carlson G.A., Prusiner S.B., Parker R (2021) **Tau aggregates are RNA-protein assemblies that mislocalize multiple nuclear speckle components** *Neuron* **109**:1675–1691 <https://doi.org/10.1016/j.neuron.2021.03.026>
78. Hernández-Ortega K., García-Esparcia P., Gil L., Lucas J.J., Ferrer I (2016) **Altered machinery of protein synthesis in Alzheimer's: from the nucleolus to the ribosome** *Brain Pathol* **26**:593–605 <https://doi.org/10.1111/bpa.12335>
79. Evans H.T., Taylor D., Kneynsberg A., Bodea L.G., Götz J (2021) **Altered ribosomal function and protein synthesis caused by tau** *Acta Neuropathol Commun* **9** <https://doi.org/10.1186/s40478-021-01208-4>
80. Mao K., Zhang G (2022) **The role of PARP1 in neurodegenerative diseases and aging** *Febs J* **289**:2013–2024 <https://doi.org/10.1111/febs.15716>
81. Ash P.E.A., Vanderweyde T.E., Youmans K.L., Apicco D.J., Wolozin B (2014) **Pathological stress granules in Alzheimer's disease** *Brain Res* **1584**:52–58 <https://doi.org/10.1016/j.brainres.2014.05.052>
82. Vanderweyde T., Yu H., Varnum M., Liu-Yesucevitz L., Citro A., Ikezu T., Duff K., Wolozin B (2012) **Contrasting pathology of the stress granule proteins TIA-1 and G3BP in tauopathies** *J Neurosci* **32**:8270–8283 <https://doi.org/10.1523/JNEUROSCI.1592-12.2012>
83. Heaton J.H., Dlakic W.M., Dlakic M., Gelehrter T.D (2001) **Identification and cDNA cloning of a novel RNA-binding protein that interacts with the cyclic nucleotide-responsive sequence in the Type-1 plasminogen activator Inhibitor mRNA** *J Biol Chem* **276**:3341–3347 <https://doi.org/10.1074/jbc.M006538200>
84. Van Dyke N., Baby J., Van Dyke M.W. (2006) **Stm1p, a ribosome-associated protein, is important for protein synthesis in *Saccharomyces cerevisiae* under nutritional stress conditions** *J Mol Biol* **358**:1023–1031 <https://doi.org/10.1016/j.jmb.2006.03.018>

85. Mateju D., Eichenberger B., Voigt F., Eglinger J., Roth G., Chao J.A (2020) **Single-molecule imaging reveals translation of mRNAs localized to stress granules** *Cell* **183**:1801–1812 <https://doi.org/10.1016/j.cell.2020.11.010>
86. Ray S., Anderson E.C (2016) **Stimulation of translation by human Unr requires cold shock domains 2 and 4, and correlates with poly(A) binding protein interaction** *Sci Rep* **6** <https://doi.org/10.1038/srep22461>
87. Mitchell S.A., Brown E.C., Coldwell M.J., Jackson R.J., Willis A.E (2001) **Protein factor requirements of the Apaf-1 internal ribosome entry segment: Roles of polypyrimidine tract binding protein and upstream of N-ras** *Mol Cell Biol* **21**:3364–3374 <https://doi.org/10.1128/mcb.21.10.3364-3374.2001>
88. Dormoy-Raclet V., Markovits J., Jacquemin-Sablon A., Jacquemin-Sablon H (2005) **Regulation of Unr expression by 5'- and 3'-untranslated regions of its mRNA through modulation of stability and IRES mediated translation** *RNA Biol* **2**:e27–35 <https://doi.org/10.4161/rna.2.3.2203>
89. Abaza I., Coll O., Patalano S., Gebauer F (2006) **Drosophila UNR is required for translational repression of male-specific lethal 2 mRNA during regulation of X-chromosome dosage compensation** *Genes Dev* **20**:380–389 <https://doi.org/10.1101/gad.371906>
90. Guo A.X., Cui J.J., Wang L.Y., Yin J.Y (2020) **The role of CSDE1 in translational reprogramming and human diseases** *Cell Commun Signal* **18** <https://doi.org/10.1186/s12964-019-0496-2>
91. Walsh D., Mohr I (2011) **Viral subversion of the host protein synthesis machinery** *Nat Rev Microbiol* **9**:860–875 <https://doi.org/10.1038/nrmicro2655>
92. Locker N., Chamond N., Sargueil B (2011) **A conserved structure within the HIV gag open reading frame that controls translation initiation directly recruits the 40S subunit and eIF3** *Nucleic Acids Res* **39**:2367–2377 <https://doi.org/10.1093/nar/gkq1118>
93. Boussadia O., Niepmann M., Créancier L., Prats A.C., Dautry F., Jacquemin-Sablon H (2003) **Unr is required in vivo for efficient initiation of translation from the internal ribosome entry sites of both rhinovirus and poliovirus** *J Virol* **77**:3353–3359 <https://doi.org/10.1128/jvi.77.6.3353-3359.2003>
94. Brugier A. *et al.* (2022) **RACK1 associates with RNA-binding proteins Vigilin and SERBP1 to facilitate dengue virus replication** *J Virol* **96** <https://doi.org/10.1128/jvi.01962-21>
95. Tollervey D., Kiss T (1997) **Function and synthesis of small nucleolar RNAs** *Curr Opin Cell Biol* **9**:337–342 [https://doi.org/10.1016/s0955-0674\(97\)80005-1](https://doi.org/10.1016/s0955-0674(97)80005-1)
96. Tafforeau L., Zorbas C., Langhendries J.L., Mullineux S.T., Stamatopoulou V., Mullier R., Wacheul L., Lafontaine D.L.J (2013) **The complexity of human ribosome biogenesis revealed by systematic nucleolar screening of pre-rRNA processing factors** *Mol Cell* **51**:539–551 <https://doi.org/10.1016/j.molcel.2013.08.011>
97. Tollervey D., Lehtonen H., Jansen R., Kern H., Hurt E.C (1993) **Temperature-sensitive mutations demonstrate roles for yeast fibrillarin in pre-rRNA processing, pre-rRNA methylation, and ribosome assembly** *Cell* **72**:443–457 [https://doi.org/10.1016/0092-8674\(93\)90120-F](https://doi.org/10.1016/0092-8674(93)90120-F)

98. Iyer-Bierhoff A., Krogh N., Tessarz P., Ruppert T., Nielsen H., Grummt I (2018) **SIRT7-dependent deacetylation of Fibrillarin controls Histone H2A methylation and rRNA synthesis during the cell cycle** *Cell Rep* **25**:2946–2954 <https://doi.org/10.1016/j.celrep.2018.11.051>
99. Herdman C., Mars J.C., Stefanovsky V.Y., Tremblay M.G., Sabourin-Felix M., Lindsay H., Robinson M.D., Moss T (2017) **A unique enhancer boundary complex on the mouse ribosomal RNA genes persists after loss of Rrn3 or UBF and the inactivation of RNA polymerase I transcription** *PLoS Genet* **13** <https://doi.org/10.1371/journal.pgen.1006899>
100. Bose T. *et al.* (2012) **Cohesin proteins promote ribosomal RNA production and protein translation in yeast and human cells** *PLoS Genet* **8** <https://doi.org/10.1371/journal.pgen.1002749>
101. Calo E., Flynn R.A., Martin L., Spitale R.C., Chang H.Y., Wysocka J (2015) **25470060 RNA helicase DDX21 coordinates transcription and ribosomal RNA processing** *Nature* **518**:249–253 <https://doi.org/10.1038/nature13923>
102. Ugrinova I., Petrova M., Chalabi-Dchar M., Bouvet P (2018) **Multifaceted Nucleolin protein and its molecular partners in oncogenesis** *Adv Protein Chem Struct Biol* **111**:133–164 <https://doi.org/10.1016/bs.apcsb.2017.08.001>
103. Hendriks I.A., Larsen S.C., Nielsen M.L (2019) **An advanced strategy for comprehensive profiling of ADP-ribosylation sites using mass spectrometry-based proteomics** *Molecular Cell Proteomics* **18**:1010–1026 <https://doi.org/10.1074/mcp.TIR119.001315>
104. Bonfiglio J.J., Fontana P., Zhang Q., Colby T., Gibbs-Seymour I., Atanassov I., Bartlett E., Zaja R., Ahel I., Matic I (2017) **Serine ADP-ribosylation depends on HPF1** *Mol Cell* **65**:932–940 <https://doi.org/10.1016/j.molcel.2017.01.003>
105. Bilan V., Selevsek N., Kistemaker H.A. V., Abplanalp J., Feurer R., Filippov D. V., Hottiger M.O (2017) **New quantitative mass spectrometry approaches reveal different ADP-ribosylation phases dependent on the levels of oxidative stress** *Molecular Cell Proteomics* **16**:949–958 <https://doi.org/10.1074/mcp.O116.065623>
106. Larsen S.C., Hendriks I.A., Lyon D., Jensen L.J., Nielsen M.L (2018) **Systems-wide analysis of Serine ADP-ribosylation reveals widespread occurrence and site-specific overlap with phosphorylation** *Cell Rep* **24**:2493–2505 <https://doi.org/10.1016/j.celrep.2018.07.083>
107. Tao Z., Gao P., Liu H.W (2009) **Studies of the expression of human poly(ADP-ribose) polymerase-1 in *Saccharomyces cerevisiae* and identification of PARP-1 substrates by yeast proteome microarray screening** *Biochemistry* **48**:11745–11754 <https://doi.org/10.1021/bi901387k>
108. Manco G., Lacerra G., Porzio E., Catara G (2022) **ADP-ribosylation post-translational modification: an overview with a focus on RNA biology and new pharmacological perspectives** *Biomolecules* **12** <https://doi.org/10.3390/biom12030443>
109. Duan Y. *et al.* (2019) **PARYlation regulates stress granule dynamics, phase separation, and neurotoxicity of disease-related RNA-binding proteins** *Cell Res* **29**:233–247 <https://doi.org/10.1038/s41422-019-0141-z>
110. Melikishvili M., Chariker J.H., Rouchka E.C., Fondufe-Mittendorf Y.N (2017) **Transcriptome-wide identification of the RNA-binding landscape of the chromatin-associated protein PARP1 reveals functions in RNA biogenesis** *Cell Discov* **3** <https://doi.org/10.1038/celldisc.2017.43>

111. Leung A.K.L (2020) **Poly(ADP-ribose): a dynamic trigger for biomolecular condensate formation** *Trends Cell Biol* **30**:370–383 <https://doi.org/10.1016/j.tcb.2020.02.002>
112. Ottoz D.S.M., Berchowitz L.E (2020) **The role of disorder in RNA binding affinity and specificity: the role of disorder in RNA binding** *Open Biol* **10** <https://doi.org/10.1098/rsob.200328>
113. Naskar A., Nayak A., Salaikumaran M.R., Vishal S.S., Gopal P.P (2023) **Phase separation and pathologic transitions of RNP condensates in neurons: implications for amyotrophic lateral sclerosis, frontotemporal dementia and other neurodegenerative disorders** *Front Mol Neurosci* **16** <https://doi.org/10.3389/fnmol.2023.1242925>
114. Lyu K., Chow E.Y.C., Mou X., Chan T.F., Kwok C.K (2021) **RNA G-quadruplexes (rG4s): genomics and biological functions** *Nucleic Acids Res* **49**:5426–5450 <https://doi.org/10.1093/nar/gkab187>
115. Banco M.T., Ferré-D'Amaré A.R (2021) **The emerging structural complexity of G-quadruplex RNAs** *RNA* **27**:390–402 <https://doi.org/10.1261/rna.078238.120>
116. Dumas L., Herviou P., Dassi E., Cammas A., Millevoi S (2021) **G-quadruplexes in RNA biology: recent advances and future directions** *Trends Biochem Sci* **46**:270–283 <https://doi.org/10.1016/j.tibs.2020.11.001>
117. Edwards A.D., Marecki J.C., Byrd A.K., Gao J., Raney K.D (2021) **G-quadruplex loops regulate PARP-1 enzymatic activation** *Nucleic Acids Res* **49**:416–431 <https://doi.org/10.1093/nar/gkaa1172>
118. Bleuzé L., Triaca V., Borreca A (2021) **FMRP-driven neuropathology in Autistic spectrum disorder and Alzheimer's disease: A losing game** *Front Mol Biosci* **8** <https://doi.org/10.3389/fmolb.2021.699613>
119. Fyke W., Velinov M (2021) **Fmr1 and autism, an intriguing connection revisited** *Genes (Basel)* **12** <https://doi.org/10.3390/genes12081218>
120. Davis J.K., Broadie K (2017) **Multifarious functions of the Fragile X mental retardation protein** *Trends Genet* **33**:703–714 <https://doi.org/10.1016/j.tig.2017.07.008>
121. Darnell J.C., Warren S.T., Darnell R.B (2004) **The Fragile X mental retardation protein, FMRP, recognizes G-quartets** *Ment Retard Dev Disabil Res Rev* **10**:49–52 <https://doi.org/10.1002/mrdd.20008>
122. Darnell J.C., Klann E (2013) **The translation of translational control by FMRP: therapeutic targets for FXS** *Nat Neurosci* **16**:1530–1536 <https://doi.org/10.1038/nn.3379>
123. Richter J.D., Zhao X (2021) **The molecular biology of FMRP: new insights into fragile X syndrome** *Nat Rev Neurosci* **22**:209–222 <https://doi.org/10.1038/s41583-021-00432-0>
124. Chen H.H., Yu H.I., Chiang W.C., Lin Y. De, Shia B.C., Tarn W.Y. (2012) **hnRNP Q regulates Cdc42-mediated neuronal morphogenesis** *Mol Cell Biol* **32**:2224–2238 <https://doi.org/10.1128/mcb.06550-11>
125. Duning K., Buck F., Barnekow A., Kremerskothen J (2008) **SYNCRIP, a component of dendritically localized mRNPs, binds to the translation regulator BC200 RNA** *J Neurochem* **105**:351–359 <https://doi.org/10.1111/j.1471-4159.2007.05138.x>

126. Khudayberdiev S., Soutschek M., Ammann I., Heinze A., Rust M.B., Baumeister S., Schratt G (2021) **The cytoplasmic SYNCRIP mRNA interactome of mammalian neurons** *RNA Biol* **18**:1252–1264 <https://doi.org/10.1080/15476286.2020.1830553>
127. Titlow J., Robertson F., Järvelin A., Ish-Horowicz D., Smith C., Gratton E., Davis I (2020) **Syncrip/hnRNP Q is required for activity-induced Msp300/Nesprin-1 expression and new synapse formation** *J Cell Biol* **219** <https://doi.org/10.1083/jcb.201903135>
128. Halstead J.M., Lin Y.Q., Durraine L., Hamilton R.S., Ball G., Neely G.G., Bellen H.J., Davis I (2014) **Syncrip/hnRNP Q influences synaptic transmission and regulates BMP signaling at the Drosophila neuromuscular synapse** *Biol Open* **3**:839–849 <https://doi.org/10.1242/bio.20149027>
129. Sapir T., Kshirsagar A., Gorelik A., Olender T., Porat Z., Scheffer I.E., Goldstein D.B., Devinsky O., Reiner O (2022) **Heterogeneous nuclear ribonucleoprotein U (HNRNPU) safeguards the developing mouse cortex** *Nat Commun* **13** <https://doi.org/10.1038/s41467-022-31752-z>
130. Lelieveld S.H. *et al.* (2016) **Meta-analysis of 2, 104 trios provides support for 10 new genes for intellectual disability** *Nat Neurosci* **19**:1194–1196 <https://doi.org/10.1038/nn.4352>
131. Balasubramanian M, Adam M.P., Mirzaa G.M., Pagon R.A., Wallace S.E., Bean L.J.H., Gripp K.W., Amemiya A (2022) **HNRNPU-related neurodevelopmental disorder** *GeneReviews® [Internet]* Seattle (WA): University of Washington, Seattle :1993–2023
132. Bramswig N.C. *et al.* (2017) **Heterozygous HNRNPU variants cause early onset epilepsy and severe intellectual disability** *Hum Genet* **136**:821–834 <https://doi.org/10.1007/s00439-017-1795-6>
133. Gillentine M.A. *et al.* (2021) **Rare deleterious mutations of HNRNP genes result in shared neurodevelopmental disorders** *Genome Med* **13** <https://doi.org/10.1186/s13073-021-00870-6>
134. Wang T. *et al.* (2020) **Large-scale targeted sequencing identifies risk genes for neurodevelopmental disorders** *Nat Commun* **11** <https://doi.org/10.1038/s41467-020-18723-y>
135. Bakkar N., Kovalik T., Lorenzini I., Spangler S., Lacoste A., Sponaugle K., Ferrante P., Argentinis E., Sattler R., Bowser R (2018) **Artificial intelligence in neurodegenerative disease research: use of IBM Watson to identify additional RNA-binding proteins altered in amyotrophic lateral sclerosis** *Acta Neuropathol* **135**:227–247 <https://doi.org/10.1007/s00401-017-1785-8>
136. Keefe A.D., Wilson D.S., Seelig B., Szostak J.W (2001) **One-step purification of recombinant proteins using a nanomolar-affinity streptavidin-binding peptide, the SBP-tag** *Protein Expr Purif* **23**:440–446 <https://doi.org/10.1006/prep.2001.1515>
137. Huang D., Camacho C. V., Setlem R., Ryu K.W., Parameswaran B., Gupta R.K., Kraus W.L (2020) **Functional interplay between histone H2B ADP-ribosylation and phosphorylation controls adipogenesis** *Mol Cell* **79**:934–949 <https://doi.org/10.1016/j.molcel.2020.08.002>
138. Chou T.C (2010) **Drug combination studies and their synergy quantification using the chou-talalay method** *Cancer Res* **70**:440–446 <https://doi.org/10.1158/0008-5472.CAN-09-1947>
139. Szklarczyk D. *et al.* (2023) **The STRING database in 2023: protein– protein association networks and functional enrichment analyses for any sequenced genome of interest** *Nucleic Acids Res* **51**:D638–D646 <https://doi.org/10.1093/nar/gkac1000>

140. Doncheva N.T., Morris J.H., Holze H., Kirsch R., Nastou K.C., Cuesta-Astro Y., Rattei T., Szklarczyk D., von Mering C., Jensen L.J. (2023) **Cytoscape stringApp 2.0: analysis and visualization of heterogeneous biological networks** *J Proteome Res* **22**:637–646 <https://doi.org/10.1021/acs.jproteome.2c00651>
141. Puchalski R.B. *et al.* (2018) **An anatomic transcriptional atlas of human glioblastoma** *Science* **360**:660–663 <https://doi.org/10.1126/science.aaf2666>
142. van de Leemput J. *et al.* (2014) **CORTECON: a temporal transcriptome analysis of in vitro human cerebral cortex development from human embryonic stem cells** *Neuron* **83**:51–68 <https://doi.org/10.1016/j.neuron.2014.05.013>
143. Fang H., Gough J (2013) **DcGO: database of domain-centric ontologies on functions, phenotypes, diseases and more** *Nucleic Acids Res* **41**:D536–544 <https://doi.org/10.1093/nar/gks1080>
144. Sutphin G.L., Mahoney J.M., Sheppard K., Walton D.O., Korstanje R (2016) **WORMHOLE: novel least diverged ortholog prediction through machine learning** *PLoS Comput Biol* **12** <https://doi.org/10.1371/journal.pcbi.1005182>
145. Kitts P.A. *et al.* (2016) **Assembly: a resource for assembled genomes at NCBI** *Nucleic Acids Res* **44**:D73–D80 <https://doi.org/10.1093/nar/gkv1226>
146. Frankish A. *et al.* (2021) **GENCODE 2021** *Nucleic Acids Res* **49**:D916–D923 <https://doi.org/10.1093/nar/gkaa1087>
147. Dobin A., Davis C.A., Schlesinger F., Drenkow J., Zaleski C., Jha S., Batut P., Chaisson M., Gingeras T.R (2013) **STAR: ultrafast universal RNA-seq aligner** *Bioinformatics* **29**:15–21 <https://doi.org/10.1093/bioinformatics/bts635>
148. Shen S., Park J.W., Lu Z.X., Lin L., Henry M.D., Wu Y.N., Zhou Q., Xing Y (2014) **rMATS: robust and flexible detection of differential alternative splicing from replicate RNA-Seq data** *Proc Natl Acad Sci U S A* **111**:E5593–E5601 <https://doi.org/10.1073/pnas.1419161111>
149. Quinlan A.R., Hall I.M (2010) **BEDTools: a flexible suite of utilities for comparing genomic features** *Bioinformatics* **26**:841–842 <https://doi.org/10.1093/bioinformatics/btq033>
150. Bray N.L., Pimentel H., Melsted P., Pachter L (2016) **Near-optimal probabilistic RNA-seq quantification** *Nat Biotechnol* **34**:525–527 <https://doi.org/10.1038/nbt.3519>
151. Sonesson C., Love M.I., Robinson M.D (2015) **Differential analyses for RNA-seq: transcript-level estimates improve gene-level inferences** *F1000Res* **4** <https://doi.org/10.12688/f1000research.7563.1>
152. Love M.I., Huber W., Anders S (2014) **Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2** *Genome Biol* **15** <https://doi.org/10.1186/s13059-014-0550-8>

Author information

Kira Breunig[†]

Children's Cancer Research Institute, UT Health San Antonio, San Antonio, USA

[†]Joint Authors

Xiufen Lei[†]

Children's Cancer Research Institute, UT Health San Antonio, San Antonio, USA

[†]Joint Authors

Mauro Montalbano[†]

Mitchell Center for Neurodegenerative Diseases, University of Texas Medical Branch, Galveston, USA, Department of Neurology, University of Texas Medical Branch, Galveston, USA

[†]Joint Authors

Gabriela DA Guardia

Centro de Oncologia Molecular, Hospital Sírio-Libanês, São Paulo, Brazil

Shiva Ostadrahimi

Children's Cancer Research Institute, UT Health San Antonio, San Antonio, USA, Department of Cell Systems and Anatomy, UT Health San Antonio, San Antonio,, USA

Victoria Alers

Children's Cancer Research Institute, UT Health San Antonio, San Antonio, USA, Department of Cell Systems and Anatomy, UT Health San Antonio, San Antonio,, USA, Department of Biochemistry and Structural Biology, UT Health San Antonio, San Antonio, USA

Adam Kosti

Children's Cancer Research Institute, UT Health San Antonio, San Antonio, USA, Department of Cell Systems and Anatomy, UT Health San Antonio, San Antonio,, USA

Jennifer Chiou

Department of Nutritional Sciences, College of Natural Sciences, University of Texas at Austin, Austin, USA

Nicole Klein

Children's Cancer Research Institute, UT Health San Antonio, San Antonio, USA

Corina Vinarov

Children's Cancer Research Institute, UT Health San Antonio, San Antonio, USA

Lily Wang

Children's Cancer Research Institute, UT Health San Antonio, San Antonio, USA

Mujia Li

Children's Cancer Research Institute, UT Health San Antonio, San Antonio, USA

Weidan Song

Laboratory of Signaling and Gene Regulation, Cecil H. and Ida Green Center for Reproductive Biology Sciences, The University of Texas Southwestern Medical Center, Dallas, USA

W Lee Kraus

Laboratory of Signaling and Gene Regulation, Cecil H. and Ida Green Center for Reproductive Biology Sciences, The University of Texas Southwestern Medical Center, Dallas, USA
ORCID iD: [0000-0002-8786-2986](https://orcid.org/0000-0002-8786-2986)

David S Libich

Children's Cancer Research Institute, UT Health San Antonio, San Antonio, USA, Department of Biochemistry and Structural Biology, UT Health San Antonio, San Antonio, USA
ORCID iD: [0000-0001-6492-2803](https://orcid.org/0000-0001-6492-2803)

Stefano Tiziani

Department of Nutritional Sciences, College of Natural Sciences, University of Texas at Austin, Austin, USA, Department of Pediatrics, Dell Medical School, University of Texas at Austin, Austin, USA, Department of Oncology, Dell Medical School, University of Texas at Austin, Austin, USA

Susan T Weintraub

Department of Biochemistry and Structural Biology, UT Health San Antonio, San Antonio, USA

Pedro AF Galante

Centro de Oncologia Molecular, Hospital Sírio-Libanês, São Paulo, Brazil

Luiz OF Penalva

Children's Cancer Research Institute, UT Health San Antonio, San Antonio, USA, Department of Cell Systems and Anatomy, UT Health San Antonio, San Antonio, USA

For correspondence: penalva@uthscsa.edu

Editors

Reviewing Editor

Paul Donlin-Asp

University of Edinburgh, Edinburgh, United Kingdom

Senior Editor

Lori Sussel

University of Colorado Anschutz Medical Campus, Aurora, United States of America

Reviewer #1 (Public review):

Summary:

Here the authors convincingly identify and characterize the SERBP1 interactome and further define its role in the nucleus, where it is associated with complexes involved in splicing, cell division, chromosome structure, and ribosome biogenesis. Many of the SERBP1-associated proteins are RNA-binding proteins and SERBP1 exerts its impact, at least in part, through these players. SERBP1 is mostly disordered but along with its associated proteins displays a preference for G4 binding and can bind to PAR and be PARylated. They present data that

strongly suggest that complexes in which SERBP1 participates are assembled through G4 or PAR binding. The authors suggest that because SERBP1 lacks traditional functional domains yet is clearly involved in distinct regulatory complexes, SERBP1 likely acts in the early steps of assembly through the recognition of interacting sites present in RNA, DNA, and proteins.

Strengths:

The data is very convincing and demonstrated through multiple approaches.

Weaknesses:

None. The authors have adequately addressed earlier reviewer concerns.

<https://doi.org/10.7554/eLife.98152.2.sa2>

Reviewer #2 (Public review):

Summary:

In this study the authors have used pull-down experiments in a cell line overexpressing tagged SERPINE1 mRNA binding protein 1 (SERBP1) followed by mass spectrometry-based proteomics, to establish its interactome. Extensive analyses are performed to connect the data to published resources. The authors attempt to connect SERBP1 to stress granules and Alzheimer's disease associated tau pathology. Based on the interactome, the authors propose a cross-talk between SERBP1 and PARP1 functions.

Strengths:

The main strength of this study lies in the extensive proteomics data analysis, and its effort to connect the data to published studies.

Weaknesses:

Support for the proposed model: While the authors propose a feedback regulatory model for SERBP1 and PARP1 function, strong evidence for PARylation modulating SERBP1 functions is lacking. PARP inhibition decreasing the amount of PARylated proteins associated with SERBP1 and likely all other PARylated proteins is expected.

Evidence from autopsy brain tissue: This study shows unexplained round, punctate staining for SERBP1 in immunohistochemistry (IHC) staining. This may be due to poor preservation of cellular structures in frozen autopsy brain tissue. SERBP1 and pTau co-staining lacks an age matched non-AD control. Most quantifications of human IHC staining and co-localization do not indicate the number of cases and what data points are shown.

The link to stress granules (SGs): G3BP1 staining indicates cytoplasmic mislocalization and perhaps aggregation pathology, but not necessarily SGs. It is not clear whether physiological transient stress granules are preserved in autopsy brain tissue. The co-localization of abundant cytoplasmic G3BP1 and SERBP1 under normal conditions does not indicate association with SGs. Stress granule proteins assemble phase-separated granules in the cytoplasm under cellular stress, whereas here it is shown that normally cytoplasmic SERBP1 has a nucleocytoplasmic distribution in the presence of H₂O₂, with no evidence for SG formation.

<https://doi.org/10.7554/eLife.98152.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer 1 (Public Review):*Summary:*

Here the authors convincingly identify and characterize the SERBP1 interactome and further define its role in the nucleus, where it is associated with complexes involved in splicing, cell division, chromosome structure, and ribosome biogenesis. Many of the SERBP1-associated proteins are RNA-binding proteins and SERBP1 exerts its impact, at least in part, through these players. SERBP1 is mostly disordered but along with its associated proteins displays a preference for G4 binding and can bind to PAR and be PARylated. They present data that strongly suggest that complexes in which SERBP1 participates are assembled through G4 or PAR binding. The authors suggest that because SERBP1 lacks traditional functional domains yet is clearly involved in distinct regulatory complexes, SERBP1 likely acts in the early steps of assembly through the recognition of interacting sites present in RNA, DNA, and proteins.

Strengths:

The data is very convincing and demonstrated through multiple approaches.

Weaknesses:

No weaknesses were identified by this reviewer.

Reviewer #2 (Public Review):*Summary:*

In this study the authors have used pull-down experiments in a cell line overexpressing tagged SERPINE1 mRNA binding protein 1 (SERBP1) followed by mass spectrometry-based proteomics, to establish its interactome. Extensive analyses are performed to connect the data to published resources. The authors attempt to connect SERBP1 to stress granules and Alzheimer's disease-associated tau pathology. Based on the interactome, the authors propose a cross-talk between SERBP1 and PARP1 functions.

Strengths:

The main strength of this study lies in the proteomics data analysis, and its effort to connect the data to published studies.

Weaknesses:

While the authors propose a feedback regulatory model for SERBP1 and PARP1 functions, strong evidence for PARylation modulating SERBP1 functions is lacking. PARP inhibition decreasing the amount of PARylated proteins associated with SERBP1 and likely all other PARylated proteins is expected. This study is also incomplete in its attempt to establish a connection to Alzheimer's disease related tauopathy. A single AD case is not sufficient, and frozen autopsy tissue shows unexplained punctate staining likely due to poor preservation of cellular structures for immunohistochemistry. There is a lack of essential demographic data, source of the tissue, brain regions shown, and whether there was an IRB protocol for the human brain tissue. The presence of phase-separated transient stress granules in an autopsy brain is unlikely, even if G3BP1 staining is present. Normally, stress granule proteins move to the cytoplasm under cellular stress, whereas SERBP1 becomes nuclear. The co-localization of abundant cytoplasmic G3BP1 and SERBP1 under normal conditions does not indicate an association with stress granules.

Reviewer #3 (Public Review):

Summary:

A survey of SERBP1-associated functions and their impact on the transcriptome upon gene depletion, as well as the identification of chemical inhibitors upon gene over-expression.

Strengths:

(1) Provides a valuable resource for the community, supported by statistical analyses.

(2) Offers a survey of different processes with correlation data, serving as a good starting point for the community to follow up.

Weaknesses:

(1) The authors provided numerous correlations on diverse topics, from cell division to RNA splicing and PARP1 association, but did not follow up their findings with experiments, offering little mechanistic insight into the actual role of SERBP1. The model in Figure 5D is entirely speculative and lacks data support in the manuscript.

Our article includes several pieces of evidence that support SERBP1's role in splicing, translation, cell division and association with PARP1. We respectfully disagree that the model in Figure 5D is speculative. The goal of our study was to generate initial evidence of SERBP1 involvement in different biological processes based on its interactome. The characterization of molecular mechanisms in all these scenarios requires a substantial amount work and will be the topic of follow up manuscripts.

(2) Following up with experiments to demonstrate that their findings are real (e.g., those related to splicing defects and the PARylation/PAR-binding association) would be beneficial. For example, whether the association between PARP1 and SERBP1 is sensitive to PAR-degrading enzymes is unclear.

We included experiments showing the interaction between endogenous SERBP1 and PARP1. Additionally, we demonstrated that SERBP1 interaction with PARP1 was disrupted when cells are treated with PARP inhibitors.

(3) They did not clearly articulate how experiments were performed. For instance, the drug screen and even the initial experiment involving the pull-down were poorly described. Many in the community may not be familiar with vectors such as pSBP or pUltra without looking up details.

We provided additional details about the vectors and expanded the description of experiments in results and figure legends.

(4) The co-staining of SERBP1 with pTau, PARP1, and G3BP1 in the brain is interesting, but it would be beneficial to follow up with immunoprecipitation in normal and patient samples to confirm the increased physical association.

Thank you for this suggestion. We performed instead a Proximity Ligation Assay (PLA) on human tissue. Data was included in Fig. 7B and C. PLA between pTau and SERBP1 confirmed interaction in AD cortices as well as SERBP1 with PARP1.

(5) The combination index of 0.7-0.9 for PJ34 + siSERBP1 is weak. Could this be due to the non-specific nature of the drug against other PARPs? Have the authors looked into this

possibility?

The combination index could be considered weak in the case of U251 cells but not in the case of U343 cells. PJ34 has been shown to be mainly a PARP-1 inhibitor. Different PJ34 concentrations and different drugs will be examined in future studies. It is worth mentioning that in a genetic screening, SERBP1 has been shown to increase sensitivity to different PARP inhibitors (PMID: 37160887). This information is included in the manuscript.

Reviewer #1 (Recommendations For The Authors):

This is a really well-done piece of research that is written very well. The data are very convincing and the conclusions are well supported. Some wording in Figures 2B and D is pixelated and hard to read. All the figure legends could benefit from being expanded but this is especially true for Figures 2, 3, 7, and 8. There is a ton of data being presented and a very limited description of what was done and what is being concluded. Some of the content may not be fully comprehended by some readers with limited descriptions.

We revised all figures to assure images are clear and their resolution is high. We expanded all figure legends to provide a better explanation of the experimental design.

Reviewer #2 (Recommendations For The Authors):

The "merged" pdf file is the same as the "article".

Individual files were uploaded this time.

The abstract should spell out acronyms, such as the name of the protein Serpine1 mRNA-binding protein 1 (SERBP1).

This was not included since the abstract has a word limit.

*"SERBP1 (Serpine1 mRNA-binding protein 1) is a unique member of this group of RBPs".
In what way is it unique?*

The text was modified to better explain SERBP1's singularities.

"RBPs containing IDRs and RGG motifs are particularly relevant in the nervous system. Their misfolding contributes to the formation of pathological protein aggregates in Alzheimer's disease (AD), Frontotemporal Lobar Dementia (FTLD), Amyotrophic Lateral Sclerosis (ALS), and Parkinson's disease (PD)" -> while TDP-43 and FUS in ALS/FTD may fit this description, it is not true for tau and amyloid-beta (AD) and alpha-synuclein (PD).

"SERBP1 is a unique RBPs containing IDRs and RGG motifs yet lacks other readily recognizable, canonical or structured RNA binding motifs. Moreover, SERBP1 has been observed by our study and others as common Tau interactor in Alzheimer's Disease (AD) brains. RBPs containing IDRs (e.g. TDP-43, FUS, hnRNPs, TIA1) have been shown self-aggregate and co-aggregate with pathogenic amyloids (Tau, A β -amyloid and α -Synuclein) in AD, Frontotemporal Lobar Dementia (FTLD), Amyotrophic Lateral Sclerosis (ALS), and Parkinson's disease (PD) and this suggest that, like other IDRs RBPs, SERBP1 contributes to RNA dysmetabolism in neurodegenerative diseases".

While the authors propose a feedback regulatory model for SERBP1 and PARP1 functions, strong evidence for PARylation modulating SERBP1 functions is lacking. The fact that PARP inhibition decreases the amount of PARylated proteins associated with SERBP1 and likely all other PARylated proteins is expected and cannot count as evidence.

We included data showing that treatment with PJ34 (PARP inhibitor) decreases SERBP1 interaction with PARP1 and G3BP1. We are currently conducting a more extensive analysis to identify SERBP1 PAR binding domain and the impact of PARP inhibition on its interactions and functions. These experiments will be included in a new manuscript.

| *A single AD case is not sufficient.*

Sorry for the poor clarity. We included in the study 6 cases from age-matched controls and 6 cases of AD. We summarize all cases demographics, and the experimental application assigned to each case in Table 1. Moreover, we included a paragraph regarding Human tissue harvesting.

| *Most western blot data are not quantified from multiple replicates, as required.*

Quantifications are now provided.

| *FTLD - frontotemporal lobar degeneration (not dementia).*

This was corrected.

| *Frozen autopsy tissue is problematic due to poor preservation. The staining presented here shows unexplained punctate staining likely due to poor preservation of cellular structures for immunohistochemistry.*

We included a paragraph regarding human tissue harvesting. We have successfully used frozen tissues in our previous studies, observing a well preserved neuronal and tissue structure (PMIDs: 32855391, 31532069 and 30367664)

| *The presence of phase-separated stress granules in tissue is controversial since these are transient structures.*

| *Normally, stress granule proteins move to the cytoplasm under cellular stress, whereas SERBP1 becomes nuclear. The co-localization of abundant (and partially overexposed) cytoplasmic G3BP1 and SERBP1 under normal conditions is not evidence for association with stress granules. Does induction of stress granule formation lead to colocalization in stress granules? The H2O2 experiment suggests otherwise.*

RBPs implicated in stress response move to stress granules when cells are exposed to stress. SERBP1 has been shown to shuttle to stress granules and nucleus in stress conditions (PMID: 24205981). Our results are in agreement.

Using co-IF, we observed some overlap between G3BP1 and SERBP1 in AD tissues. As shown in Fig. S6A and B, 50% of stress granules overlap with SERBP1 signal. On the contrary, it is hard to assess their relationship in aged-matched control brains where stress granules form and accumulate with a lower rate than in AD. SERBP1 is not very abundant in normal brains. It is known that RNA-Binding Proteins aggregation and/or dysfunctional LLPS dysregulate stress granules formation and accumulation in AD and other proteinopathies (PMIDs 30853299, 27256390 and 31911437). However, it is too early to determine the role of SERBP1 and its contribution to stress granules formation and accumulation. We will examine this topic in future studies.

| *There is a lack of essential demographics data (age, clinical diagnosis, path diagnosis, co-pathologies, Braak stage, etc.), source of the tissue (what brain bank?), brain regions*

shown, and whether there was informed consent for the collection and use of human brain tissue.

We included the information requested in materials and methods section.

Reviewer #3 (Recommendations For The Authors):

The authors need to better explain their experimental rationale and approach in the main text, not just in the supplementary materials.

We have extensively revised the text to provide a better description of experiments in the results section and figure legends.

<https://doi.org/10.7554/eLife.98152.2.sa0>